Title: A study on how individuals of differing personality types are impacted by a cyber attack

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Abstract

This final year project investigates how individuals with differing personality types are affected by cyber attacks. The study aims to utilize the Five-Factor personality model alongside the IES-R: Impact of Events Scale-Revised alongside a number of other questions within a questionnaire to provide a dataset that can be analyzed to evidence differing responses between individuals who take the questionnaire.

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Chapter 1: Introduction

1.1 The Human Impact of Cybersecurity Incidents

As technology continues to develop and shape multiple aspects of an individual's life, people have become increasingly dependent on online systems for many parts of their life, relying on them for personal, financial or professional interactions with others. However, this increasing reliance on these set of technologies have exposed them to a growing range of cyber threats, such as: Phishing attacks, ransomware, identity theft and data breaches have more frequently affected the average person worldwide and while cybersecurity research primarily focuses on prevention strategies and technical vulnerabilities, there is growing recognition for cyber incidents and their psychological effects on a target.

1.2 Project aim and objectives

This study aims to investigate how individuals with different personality types may respond to a cyber threat or how their reaction will differ if they are the target of a cyberattack, identifying emotional responses triggered by cyber incidents and determining how personality traits influence an individual's reactions. To achieve this, the study will use an anonymous survey to gather the experiences of multiple people that have been the target of a cyber attack. The data gathered will then be analyzed to explore the relationship between personality and the psychological impact of cyber attacks.

The study will seek to accomplish a number of key research objectives. Firstly, it will need to have a developed understanding of prior texts and works on the subject matter, to do this a literature review will be carried for a wide range of academic literature, recorded statistics, including survey-based research that examines the emotional response and experiences of targets of cyber attacks.

Secondly, this project seeks to explore the role of a person's personality within cybersecurity and how it can shape their individual responses to cyberincidents. Building upon the work presented in *Emotional Reactions to Cybersecurity Breach Situations: Scenario-Based Survey* by Budimir et al (2021.). While the original study utilized models such as the Depression Anxiety Stress Scales (DASS-21), the ego resilience scale and aggression measures to evaluate the emotional reaction to cyber attacks, this study will instead use the Impact of Events Scale - Revised (IES-R) as an instrument to measure the impact of a cyber incident within the questionnaire.

1.3 Legal, ethical, professional, social issues

From a legal perspective, the primary concern of this study relates to the handling and storage of participant data that is gathered through the survey. While the research involves gathering an individual's responses on how a cyberattack affected them, it is essential that the collected data, even if anonymous, is treated as confidential and handled in accordance with the relevant data protection regulations. As the study is being carried out in the United Kingdom, this would include compliance with the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018, these require for data to be stored securely and to be used only for the purposes outlined and intended for the project.

Ethically, the study should remain transparent and the individual taking the questionnaire should be fully informed of what the information they are submitting to the study will be used for and their right to withdraw before agreeing to submit their response. Prior to the questionnaire, an ethics sheet will be provided to ensure the participant understands fully what they are consenting their response to be used for alongside their right to withdraw their response from the questionnaire, because the information gathered will be fully anonymous, there will be no way to withdraw once a response is submitted.

Additionally, there are the professional issues that the study will have to address. As it is an academic project, it is important that all research is conducted with integrity, accuracy and objectivity, ensuring that the data collected is collected and analysed responsibility and avoiding bias, findings must be accurately represented within the final report. In addition to this, the sources used for the literature review must be rigid and reliable and properly cited.

The study should also acknowledge social considerations regarding the research, while the questionnaire utilizes the Big Five personality framework to categorize individual personality traits and how these may influence responses to cyberattacks. However, while it is a widely used psychological instrument to this day, some critics argue that it is a dated model for understanding human personality.

1.4 The report

The introduction will provide an overview of the research topic, outlining the significance of growing cybersecurity incidents and their impact on an individual's personality, this chapter will introduce and provide a background to the study, defining the project's aims and objectives while also discussing the legal, social, ethical and professional considerations with conducting the survey.

Following the introduction, the literature review will provide critical examination of prior academic texts that are relevant to the project. This chapter will explore previous studies around the subject matter, alongside researching and introducing personality and impact scales such as the Big Five personality model (McCrae & John, 1992) and the Impact of Events Scale (IES-R) (Weiss & Marmar, 1997) which will then be discussed to identify possible knowledge gaps within modern research.

Primary research will outline the design and implementation of the research conducted for the study, describing the project's research strategy, participant recruitment process and questionnaire design, this will include the chosen models used within the survey to evaluate personality and the emotional response to a cyberincident.

The data analysis will present the processing of data collected through the primary research phase, statistical results and observed patterns will be examined to identify the relationship between personality traits and emotional response within the respondent's submitted response to the survey.

Following the analysis, there will be a chapter dedicated to discussing the results from it, how it can be interpreted in relation to the project's aims and objectives or prior texts around the subject matter. Finally, the conclusion will summarise the key findings that were found through the study project, then evaluate whether the project's targets were successfully achieved. Overall, this chapter will highlight the contributions of the study, it will also provide reflection on the work done and its implications for cybersecurity, while also providing recommendations for future research on the topic.

Chapter 2: Literature Review

2.1 Introduction

This chapter will be used to present a critical review of already existing academic literature papers that can be utilized to build a foundation for my study. The literature review will be used to formally examine and evaluate prior scholarly research, primarily it will focus on cyber attacks and the psychological impact on individuals, focusing on the aspect that individual's differing personalities may influence the response to such incidents. By examining popular frameworks such as the Big Five Personality Model (McCrae & John, 1992) and other current theories, this review will provide an overview of the topic.

2.2 Cyber Attacks and their impact

Due to increasing reliance on digital technology and growing amount of information, an individual's exposure to cyber threats has grown increasingly through each passing year (Hassan et al, 2025), presenting not only a technical issue, but also a psychological and social concern. These types of cyber incidents such as ransomware attacks, data breaches and system intrusions are prone to disrupting an individual's daily life, generating emotional and behavioral consequences for the victims (Kowalski et al, 2014).

Table 5. Compound Annual Growth Rate (CAGR) of Cybersecurity Incidents (2020-2023)

Incident Type	CAGR
Malware	8.2%
Phishing	17.1%
DDoS	16.9%
Ransomware	21.3%
Data Breaches	28.3%

Table 2.1: Compound Annual Growth Rate (CAGR) of Cybersecurity Incidents (Hassan et al, 2020-2023)

This increase in cybersecurity incidents can be seen demonstrated in Table 2.1, where consistent year-on-year growth across multiple attack types is documented. Data breaches are reported to have increased at a compound annual rate (CAGR) of 28.3% between three years, granting it the highest rate of growth while ransomware and DDoS among other attacks show similar significant annual increases (Hassan et al, 2025). These findings indicate a threat landscape that will increase the chances of an individual experiencing a cyber attack and the associated impacts on their personality.

Moving onto other texts about the impact of cyber attacks on an individual, such as significantly increased cortisol levels alongside feelings of personal security and impact on terror threat perception (Canetti et al, 2017), further evidencing the psychological impacts of a cyber attack on an individual and while loosely connected it also proves that cyber attacks have a further impact regarding an individual via increasing their cortisol level. This response would differ from person to person.

2.3 The emotional impact on individuals

Cyber Attacks are evidenced to generate psychological consequences, the intensity of these reactions however varies considerably between individuals, implying that a person's own personality will shift the emotional response to the incident. Targets of these attacks primarily report emotions such as anxiety, fear or anger, these feelings are recorded to be more extreme regarding the type of attack it was as well, Budimir, Fontaine and Roesch (2021) found that cyber attacks that compromised an individual's social network account generated a stronger emotional reaction than breaches where an individual's email account was compromised.

Further study on these texts indicates that cyber attacks can influence longer-term behavioral and psychological issues, for example victims of an incident have shown to develop reduced trust in online platforms, alongside increasing their wariness or otherwise avoidant behaviours when interacting with digital technologies (Budimir et al, 2021). Cyberbullying texts specifically have shown comparable emotional patterns, where the targets of this type of attack reported experiencing lower self-confidence, higher anxiety and emotional harm (Kowalski et al, 2014).

Importantly, these articles highlight that individuals react to the emotional impact from a cyberattack with varied responses, according to a UK government report from the Home Office (2025) the response to ransomware attacks varied between the thirty-nine interviewees, they experienced different levels of emotional distress, responded with different behaviours to cope and had differing, varying long-term impact after the attack. Additionally, Budimir et al. (2021) further evidences the shift in results that an individual's personality may cause on the impact from a cyber incident. Both of these texts highlight and demonstrate the possible further research that could be carried out for the study.

Both Budimir et al (2021) and Kowalski et al (2014)'s findings contribute to providing a clearer understanding of the emotional consequences that comes from cyber incidents, however the two differ significantly in scope, the methodology utilized and the focus of their studies. Kowalski et al (2014) primarily focus their studies on examining cyberbullying as a form of online victimisation, highlighting the consistent emotional impact and emotions experienced by the targets such as anxiety or reduced self-esteem, yet the focus of these studies is primarily on the behavioural and emotional outcome regarding the targets of cyberbullying, holding limited consideration for the underlying individual differences between the targets such as personality traits. On the other hand, Budimir et al (2021) focus on adopting a more psychologically integrated approach, incorporating personality frameworks such as the Big Five model to examine how individual traits shape the emotional response to cyber incidents.

Both of these studies identify a similar set of outcomes with the respondents showcasing feelings of emotional distress such as anxiety or reduced trust, however Budimir et Al's study extends this viewpoint by suggesting that these responses are not uniform across individuals, instead focusing on how the response may vary according to personality structure. Both studies have limited ecological validity, as Kowalski et al (2014) focuses primarily on cyberbullying and its targets, limiting the results on other forms of cyber incidents, while Budimir et al (2021) has studies that rely on scenario based questionnaires, which may not truly reflect the emotional impact of the cyber incident on the target as they could be influenced by self report bias. Taken together, these highlight a gap in literature that regards the integration of personality traits and psychological instruments into cyberattack response research.

2.4 Personality and Impact Models Outside of Cybersecurity

One of the primary texts that was analysed for this study was *Emotional Reactions to Cybersecurity Breach Situations: Scenario-Based Survey Study* by Budimir et al (2021), which examined the emotional dimensions associated with cybersecurity breaches and how characteristics of personality relate to the emotional impact of a cyber attack. To do this, the study applied two well established and broad personality models, these being The Big Five personality model and the resilient/overcontrolled/undercontrolled personality type model.

The Big Five framework categorizes an individual's personality into five different traits, these being extraversion, agreeableness, conscientiousness, neuroticism and openness, which are widely recognised to impact regulatory and stress responses from an individual (McCrae & John, 1992), the study by Budimir et al (2021) specifically used the IPIP-50 instrument to measure each personality factor. On the other hand the personality type model categorizes between resilient, overcontrolled and undercontrolled, focusing on this enables the identification of differing coping styles when the individual is faced with stressful events (Robins et al, 1996).

An impact model that is extensively used outside of cybersecurity is the Impact of Events Scale (IES-R), it is a 22-item self-report measure and validated psychological instrument which is utilized to evaluate an individual's distress caused by traumatic events, it is a revised version of an older model, the 15-item IES by Horowitz, Wilner, & Alvarez, 1979. The IES-R categorizes trauma across three symptoms, intrusion, avoidance and hyperarousal.

Intrusion refers to unwanted thoughts or memories of an event resurfacing even when unwanted, avoidance meanwhile focuses on the individual's attempt to suppress the thoughts and reminders of the experience and hyperarousal includes heightened alertness or emotional sensitivity. Aligning with recognised stress and trauma responses, these dimensions make the IES-R a widely recognized instrument used within psychological and behavioural research.

2.5 Personality and Impact Models within Cybersecurity

When both of these personality frameworks were utilized, Budimir et al (2021) was able to find a correlation between an individual's personality characteristics and their responses to cyber incidents, providing support for further research to adopt the same or similar frameworks to investigate the psychological response to cyber attacks.

As well as these models, Budimir et al (2021) utilized differing scales and questionnaires to analyze the impact of each cyber incident, such as the Depression, Anxiety and Stress-Scale 21 item (DASS-21) which assesses the internalizing problems of an individual (Lovibond & Lovibond, 1995), being a key feature of the overcontrolling personality type. A questionnaire that was used in this study was the Short-Form Buss-Perry Aggression Questionnaire, which assesses a key feature of the undercontrolled personality type being aggression, (Bryant & Smith, 2001). The third scale used was the Ego Resilience Scale which assesses the key resilient feature of a personality type from an individual (Alessandri et al, 2015).

The scale that will be used for this study to differ it from previous texts is the Impact of Events Scale-Revised (IES-R), it is a validated psychological instrument that was designed to measure the distress following traumatic and stressful experiences, (Weiss & Marmar, 1997). This is a tool that has not been widely utilized within existing Cybersecurity research, this study will use the IES-R 22-item instrument to expand on prior research.

Utilizing these frameworks within my own study will provide a comprehensive approach to understanding the human side of cyber attacks. Personality models such as the Big Five can allocate a person's traits across five different categories, while impact scales such as the IES-R measure how strongly the impact of the cyberattack affected their response. Integrating these approaches will support a viewpoint that is supported by widely used psychological models.

2.6 Conclusion

Despite the growing research that examines the psychological consequences of cyber attacks, the field remains less developed in comparison to other psychological fields. Existing studies that were researched for this literature review have consistently demonstrated that cyber attacks do generate a significant amount of emotional distress, including the feelings of anxiety, fear and reduced trust in digital environments (Budimir et al, 2021; Kowalski et al, 2014). However, many of the works regarding the topic treat the targets of cyber attacks as if they were a homogenous group, this limits the integration of already established instruments such as the Big Five within cybersecurity.

Where personality traits are considered, findings are often derived from scenario based studies or self reporting from the target, which may not accurately reflect the emotional response experienced for cyber incidents. Additionally, valid psychological instruments that are widely used within trauma research such as the Impact of Events Scale - Revised (IES-R) face limited application within cybersecurity fields and studies, resulting in a clear gap of understanding on how the personality traits derived from the Big Five influence the intensity of the emotional response to a cyber incident. This study aims to address this gap in knowledge through the integration of the Big Five personality framework and the IES-R to systematically examine how personality traits possibly affect the emotional response to cyber attack experiences.

To conclude, for this study extensive research was conducted to understand the increasing growth of cyber incidents alongside the psychological impact that individuals face due to cyber attacks and how their differing personalities can cause the response to an attack to shift between person to person. Alongside this research was also done on existing personality frameworks and impact scales so the study can both utilize valid tools and attempt to use instruments that have not been utilized before to see if older findings will be confirmed.

Chapter 3: Primary Research

3.1 Introduction

This chapter outlines the primary research that will be conducted for this study, the previous chapter reviewed existing academic literature, this chapter will shift focus onto the investigation that will be undertaken to further build on previous works.

For primary research, as stated in the previous chapter, the study wants to build off of existing work and research such as what was produced by Budimir et al. (2021), which while examined emotional reactions to cybersecurity breaches using psychological instruments with proven validity, this study aims to adopt alternative methodological direction by utilising the IES-R to assess emotional responses to cyber attacks, while utilizing the Big Five personality framework to assess personality.

The chapter will also present the questionnaire that was used to gather empirical data, detailing the research design, the participant recruitment process and questionnaire structure. This will include how personality traits and emotional response were measured within it. It will also outline the ethical considerations, including informed consent, confidentiality and adherence to data protection regulations that the research was responsibly conducted under alongside academic standards.

3.2 Research Design

This study will primarily follow a quantitative focused approach for the sake of research, an approach that will be supported by a small number of qualitative research added within the questionnaire. A quantitative framework was favoured due to the research's primary objective of measuring the relationship between personality traits and impact, as the Big Five Model and the IES-R have already established questions for surveys, it allows for the quantitative data gathered from responses to be standardised and statistically analysed, making it possible to identify patterns, trends and potential correlation between participants (Creswell & Creswell, 2018). While the primary focus is to gather quantitative data, a number of open-ended questions were added to the survey to gather brief qualitative insights that could further explain participant reasoning, providing additional context to findings.

Following this, the study will be employing a survey-based design research methodology, the study will be using an online questionnaire as the data collection instrument. This type of methodology is widely used across behavioural and psychological research as it allows for the systematic collection of standardised data from large data groups while also ensuring consistency within these responses (Creswell & Creswell, 2018). This research methodology was favoured due to the efficient comparison of psychological variables and emotional responses associated with cybersecurity incidents, while also minimising researcher influence during data collection (Bryman, 2016).

3.3 Research Tools

The primary research tool utilized for the study was an online questionnaire which was designed to collect structured data relating to participant personality traits and emotional response regarding the respondent's personal feelings on a cyber incident that had affected them in the past. Google Forms was chosen as the platform to distribute the questionnaire as it offers the chance to utilize Google's secure servers to store the data, is easily accessible to most individuals as most people are already owners of a google account and it allows for complete respondent anonymity.

For the survey, complete anonymity was preferred to encourage honest participant responses when discussing personal events that could have prompted potentially sensitive emotional reactions. Previous research demonstrates that anonymous surveys reduce social desirability bias, where on the other hand participants may alter their responses to present themselves more favourably. (Joinson, 1999). Another reason that complete anonymity was chosen was to improve compliance with the GDPR, as by eliminating the ability to identify individual respondents within the data set reduces privacy risks by removing personally identifiable data from the regulation's scope.

The survey was structured to first gather respondent information required to contextualize participant responses. This was chosen for the sake of allowing efficient data gathering while comparing responses from one another. The quantitative questions within this section were inspired by the Cybersecurity GRID Questionnaire (Budimir et al, 2021), as it provides a valid outline for the measurement of emotional reactions within cybersecurity scenarios, improving the reliability of collected data. At the end of this section, there are a few optional qualitative questions, which were kept optional so the respondent was not forced to discuss personal subjects, these questions were developed by building off of the guidance outlined by Mae (2024) on the design of qualitative research questionnaires.

Following this section, the survey incorporated personality assessment through the use of the IPIP Big Five model. This section made use of the ten-item scale derived from the 50-item International Personality Item Pool (IPIP) representation of the Big Five personality traits (Goldberg et al., 2006), the abbreviated scale was used to enable for efficient data gathering while also reducing the length of the survey, which then lowered survey fatigue and to maintain participant engagement. Collecting personality data allowed for respondent traits to be analysed along with their response to the following section which could then be compared with the respondent information gathered through the IES-R (Weiss & Marmar, 1997) which was fully implemented in the final section of the survey. This was done to provide a valid measure of the severity of the impact through an instrument that has previous usage in psychological fields, but has not been as widely used within the cybersecurity field.

The IES-R items were scored on a 1-5 Likert scale rather than the original 0-4 format like how Big Five IPIP scale the study is utilizing. This modification will increase the absolute total scores, but will not affect the relative comparisons or statistical relationships between variables.

3.4 Participants

The Participants for this study were recruited through the use of a voluntary sampling approach. This was done as the survey would be shared on multiple various online platforms, including forums and survey-sharing websites such as Survey Circle. This approach was chosen to reach a diverse population of adults that had experienced cyber incidents before, while also being practical and efficient as the survey would allow a broad participant pool without geographical limitations as well as ensuring full participant anonymity.

Participation for the survey was completely voluntary and no incentives were offered, respondents were informed that all their data would remain anonymous. No personally identifiable data such as names or IP addresses were collected at any stage of the questionnaire, which ensures full compliance with established data protection principles, such as the GDPR. However, once a response was submitted the respondents could not withdraw it, this was made clear in a provided ethics information document provided within the survey.

An article that presented the desired sample size in this study was the UK Government's qualitative research on ransomware victims, *The experiences and impacts of ransomware attacks on individuals and organisations*, which collected the responses from a total of 39 respondents through interviews with the affected individuals and organisations. (Home Office, 2025). The study demonstrated the value of capturing personal perspectives on ransomware impacts, which inspired the addition of qualitative questions within the study's survey. Drawing from this, the study aims for a minimum of 40 full survey responses, as it would provide a balanced need of varied perspectives while also acknowledging the possible difficulties regarding the practicality of online recruitment.

3.5 Summary

This chapter outlined the design and implementation of the primary research that will be conducted for the study, detailing that it primarily adopted a quantitative approach to gather data on and investigate the relationship between how an individual's personality will affect their response to a cyber attack. The research design justified the use of survey-based methodology, enabling standardised data collection that utilized valid psychological instruments such as the Big Five personality model and the Impact of Events Scale revised (IES-R). The development of this online questionnaire allowed for participants to be able to respond to the survey without any limitations while also remaining completely anonymous.

Chapter 4: Data Analysis

4.1 Introduction

This chapter will be used to present the statistical analyses that were conducted to examine the respondents' result to the questionnaire, focusing on how this can evidence the relationship between personality traits and individual's psychological reactions to cybersecurity incidents. Data analysis was performed utilizing IBM SPSS statistics and these were conducted in several stages. To start, descriptive statistics were calculated to summarise participant responses and provide a general overview of the relationship between personality and cyberattacks within the sample of forty respondents. Second, a reliability analysis was conducted to assess and ensure the internal consistency of the psychological scales (IPIP Big Five and IES-R) that were utilized within the study. Thirdly, a principal component analysis (PCA) was conducted to identify the underlying components within participants' reported emotional reactions to cybersecurity incidents.

4.2 Quantitative Results

Descriptive Statistics

Descriptive statistics were calculated for all cyber-attack reaction items, Big Five personality traits and the IES-R measures. Means and standard deviations are presented in Table 4.1. Utilizing IBM SPSS, the table was produced via the analyze function, onto descriptive statistics and descriptives. Where likert items such as the Big Five traits and IES-R scores were all selected as variables. The following were also added through options: Mean, std. Deviation and a Minimum and Maximum.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Frequent Target of Cyber Attacks	40	1	5	2.68	1.163
Were they impacted by the attack	40	1	5	3.25	1.276
Personality change after attack	40	1	5	2.75	1.463
Would training have made a difference	40	1	5	3.47	1.467
Did they feel secure at the time	40	1	5	3.60	1.150
Did the individual feel it happened because they did something wrong?	40	1	5	2.78	1.349
Did the user feel their trust was betrayed?	40	1	5	2.93	1.526
Are they worried similar situations may happen in the future?	40	1	5	3.18	1.448
Did you feel you could do anything about the situation?	40	1	5	2.80	1.381
Were you afraid your security and the security of people close to you could be jeopardized?	40	1	5	3.18	1.466
Extraversion Scaling	40	1.5	5.0	3.268	.9657
Agreeableness Scaling	40	2.1	5.0	3.860	.7264
Conscientiousness Scaling	40	1.0	5.0	3.538	.8643
Emotional Stability Scaling	40	1.4	5.0	3.440	.9951
Imaginative Scaling	40	1.4	4.9	3.548	.7197
IES-R Intrusion	40	8	38	21.23	8.475
IES-R Avoidance	40	8	36	20.88	7.697
IES-R Hyperarousal	40	6	28	16.70	6.354
IES-R Total	40	22	102	58.80	21.255
Valid N (listwise)	40				

Table 4.1: Descriptive Statistics Table

With the mean representing the average value of each of the responses and the standard deviation (SD) displaying the variation around this average, the two can be used as evidence to prove whether a set of responses is consistent or not. Low SD evidence consistent responses, a higher SD scoring evidences the result being spread out.

Overall, the participants reported a modest level of impact following the cybersecurity incidents they experienced. The perceived impact of the attack showed a mean score of M = 3.25 (SD = 1.28), which shows that participants generally experienced a noticeable response to the incident, however experienced varying consequences. Additionally, the feelings of security reported by respondents were relatively higher, scoring M = 3.60 and SD = 1.15, these results evidence that respondents initially felt secure before the cyber incident.

The results also evidence that respondents felt elevated concern for the possibility of future cyberincidents, M = 3.18 and SD = 1.45, these scores for mean and standard deviation were similar to the fear respondents felt that the security of people close to them could be jeopardized by the attack, M = 3.18, SD = 1.47. In contrast, the self blaming responses were lower, M = 2.78, SD = 1.35, showing how participants generally did not feel like they were the responsible cause of the incident.

Personality trait scores fell within expected mid-range for the five-point scale that was used. Agreeableness demonstrated the highest mean, M = 3.86, SD = 0.73, where on the other hand extraversion showed greater variation M = 3.27, SD = 0.97.

IES-R scores suggested that respondents experienced moderate psychological distress due to the cyber incidents. The total for the IES-R scores were M = 58.50, SD = 21.26, displaying that on average participants felt a moderate to high level of distress, with observed scores ranging from 22 to 102. From the subscales, intrusion scored M = 21.23, SD = 8.40 and avoidance scored M = 20.88, SD = 7.70, both which had a higher rating than hyperarousal, M = 16.70, SD = 6.35, which implies respondents felt a higher level of intrusive thoughts and avoidance behaviours.

Reliability Analysis

The consistency of the personality and trauma-based responses were assessed using Cronbach's alpha (α) coefficients, this was carried out to ensure that the items were valid for use in analysis. Within IBM SPSS, these results were produced by using the analyze, scale and reliability analysis. To produce α, all of the individual questions and their responses for a certain item were moved into the " Item " box, where the model was Alpha and in the Statistics options scale if item is deleted, items and correlations were chosen as options. To calculate α, Cronbach's Alpha, the number of items is divided by the average variance added to the number of items subtracted by one then multiplied by average covariance between items. Essentially, this means that the alpha increases when items correlate strongly to one another and it decreases when items behave differently.

All of the scales that were used for the study are demonstrated to stand at either acceptable or excellent reliability with the responses gathered from the questionnaire. Extraversion showed an excellent internal consistency ($\alpha = .903$), while agreeableness ($\alpha = .850$), emotional stability ($\alpha = .852$) and imaginative scaling ($\alpha = .824$) demonstrated strong reliability. Conscientiousness on the other hand exhibited an acceptable level of reliability ($\alpha = .730$), all of these results are consistent with the expectations of a short-form questionnaire.

After this, the reliability of the IES-R scale with our data set was analysed, where it demonstrated an excellent internal consistency ($\alpha = .943$), showcasing a high level of coherence amongst the gathered responses to measure the impact of the cyber incidents. Overall, item total correlations were all above .30, which indicated that all of the instruments for the study were reliably measured and utilized for their intended purpose.

Principal Component Analysis of Cyber-Attack Reactions

To further analyze the quantitative data, a Principal Component Analysis (PCA) was carried out through SPSS via Analyze, Dimension Reduction and then Factor. This was done to identify the dimensions that could be found within the ten cyber incident related questions at the start of the questionnaire, where the PCA was used to reduce the data into a smaller number of components to explain the variance within the responses of participants. Varimax rotation was applied to improve the interpretability of the results by maximizing the highest loadings, doing this produced a clear separation between the resulting components. The reason that Varimax rotation was chosen over alternative options was because the components were assumed to be conceptually independent instead of strongly correlated, the resulting Table 4.2 is shown below, portraying the two component outcome of the PCA alongside the distinct dimensions of the reactions to cyberincidents.

	Component	
	Perceived Threat and Emotional Impact	Perceived Control and Preparation
Are they worried similar situations may happen in the future?	.893	
Were you afraid your security and the security of people close to you could be jeopardized?	.823	
Did the user feel their trust was betrayed?	.663	
Did the individual feel it happened because they did something wrong?	.559	.505
Would training have made a difference		.785
Were they impacted by the attack	.549	.688
Did you feel you could do anything about the situation?		.683
Personality change after attack	.538	.621
Frequent Target of Cyber Attacks		.565
Did they feel secure at the time		.449

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

Table 4.2: Principal Component Analysis (PCA) Results

The first component was labelled the Perceived Threat and Emotional Impact, as the loading from these questions within the questionnaire were primarily related to the respondent's worry about future attacks and perceived risk for the respondent or the people close to them, this is also accompanied by feelings of betrayal and overall emotional impact due to the cyber incident. This component is labelled this way to represent the participants' emotional and threat-based feelings on the cyberattack.

The second component on the table was named the Perceived Control and Preparation, which included weight from the questions within the questionnaire that are related to the belief about training and its effectiveness against cyberattacks, respondents' perceived security at the time of the incident, their perceived ability to respond to the attack alongside the frequency of being targeted. The questions in this category reflect a respondent's self-evaluation of preparedness, control and coping capacity.

A number of the questions within the PCA table hold a shared weight across both of the components, the questions that were found with shared loading regard an individual's perceptions of personal responsibility and their reported feelings of overall impact, which suggests an overlap between the emotional distress that is felt by the participant and the respondents' interpretation of events of the incident. This discovery pushes forward evidence regarding how a person's emotional reaction to cyberincidents may be closely linked with an individuals' belief about control and responsibility.

The two components that make up the table support the perspective that cybersecurity incidents bring up both emotional threat responses and an individual's personal view of perceived control. This provides the structure for possible future analyses on the subject matter, ones that could possibly focus on examining the relationship between how personality traits affect one's own evaluation of control and their emotional threat response to cyberincidents.

Multiple Regression Analysis of Cyber-Attack Reactions

For the study, a multiple linear regression analysis was conducted through IBM SPSS, allowing the examination of whether Big Five personality traits or the Perceived Control component derived from the PCA predicted the emotional impact and psychological response to a cyberincident. The way that multiple linear regression calculates an estimation of the relationship between several independent variables and a singular dependent variable is through the use of a linear model that utilizes a weighted combination of predictors which then minimises the difference between observed and predicted values. This allows for the unique contribution of each predictor to be assessed while controlling for the others.

In this study, the dependent variables were chosen to be the IES-R Total score and its subscales: Intrusion, avoidance and hyperarousal, while the predictor variables contained the IPIP Big Five traits and Emotional Impact and Perceived Control components which were derived from the PCA. Using the enter method within SPSS, all predictors were chosen at the same time via Analyze, Regression, Linear. In addition to this, within the statistics option the following were chosen: Estimates, Model Fit and Confidence Intervals.

Table 4.3: Regression Analysis with IES-R Total as Dependent Variable

Predictor	B	SE B	β	t	p
Extraversion	-0.42	3.53	-0.02	-0.12	0.91
Agreeableness	-7.84	4.49	-0.27	-1.74	0.09
Conscientiousness	-2.77	4.20	-0.11	-0.66	0.52
Emotional Stability	-1.69	3.67	-0.08	-0.46	0.65
Imaginative Scaling	3.90	4.65	0.13	0.84	0.41
Emotional Impact	13.15	3.17	0.62	4.15	< 0.001
Perceived Control	6.97	2.92	0.33	2.39	0.02
Constant	92.19	23.13		3.99	< .001

Table 4.4: Regression Analysis with IES-R Intrusion as Dependent Variable

Predictor	B	SE B	β	t	p
Extraversion	-0.16	1.41	-0.19	-0.16	0.91
Agreeableness	-1.74	1.80	-0.15	-0.97	0.34
Conscientiousness	-1.43	1.68	-0.15	-0.85	0.40
Emotional Stability	-0.84	1.47	-0.99	-0.57	0.57
Imaginative Scaling	1.99	1.86	0.17	1.07	0.29
Emotional Impact	5.08	1.27	0.60	4.00	<0.00
Perceived Control	2.42	1.17	0.29	2.07	0.47
Constant	29.37	9.26		3.17	0.003

Table 4.5: Regression Analysis with IES-R Avoidance as Dependent Variable

Predictor	B	SE B	β	t	p
Extraversion	-0.35	1.46	-0.44	-0.24	0.81
Agreeableness	-3.37	1.86	-0.32	-1.82	0.08
Conscientiousness	-0.14	1.73	-0.06	-0.08	0.94
Emotional Stability	-0.80	1.52	-0.10	-0.53	0.60
Imaginative Scaling	0.92	1.92	0.09	0.49	0.64
Emotional Impact	3.65	1.31	0.47	2.79	0.01
Perceived Control	2.25	1.21	0.29	1.86	0.07
Constant	35.01	9.55		3.67	<.001

Table 4.6: Regression Analysis with IES-R Hyperarousal as Dependent Variable

Predictor	B	SE B	β	t	p
Extraversion	0.09	0.98	0.14	0.91	0.93
Agreeableness	-2.73	1.25	-0.31	-2.17	0.04
Conscientiousness	-1.20	1.17	-0.16	-1.02	0.31
Emotional Stability	-0.54	1.02	-0.01	-0.05	0.96
Imaginative Scaling	1.00	1.30	0.11	0.77	0.45
Emotional Impact	4.43	0.88	0.70	5.01	<.001
Perceived Control	2.31	0.81	0.36	2.84	0.01
Constant	27.80	6.45		4.31	<.001

Key:

B - Unstandardized Coefficients

SE - Standard Error

β - Standardized Coefficients Beta

t - t value / Contribution to dependent variable

p - Sig / Statistical reliability

Table 4.7: Regression Model Summary for IES-R Total and Subscales

DV	R	R^2	F (df1, df2)	Significant Predictors
IES-R Total	0.71	0.50	4.53*	Emotional Impact, Perceived Control
Intrusion	0.70	0.49	4.47*	Emotional Impact
Avoidance	0.59	0.35	2.44*	Emotional Impact
Hyperarousal	0.750	0.56	5.88*	Emotional Impact, Perceived Control

R - Multiple Correlation Coefficient

R^2 - Proportion of variance in the DV

F (df1 and df2) - Overall model significance test

The regression model predicted that the IES-R total scores were statistically significant, displaying a substantial proportion of variance within the DV as shown by how $R^2 = 0.49$, $F = 4.53$ and $P < 0.05$, where emotional impact ($\beta = 0.62$, $p < 0.001$) and perceived control ($\beta = 0.33$, $p < 0.02$) came out as the most significant predictors, indicating that the greater the emotional response to an incident the greater the level of psychological distress was as well as evidencing that an individual's sense of control or responsibility over their security also increased the level of psychological distress. On the other hand, the regression model pointed to neither of the Big Five traits significantly influencing the distress caused by a cyber incident as all of them showed $p > 0.5$, indicating a lack of meaningful contribution.

Alongside this, applying the regression model onto the IES-R subscales revealed a consistent pattern where Emotional Impact still remained one of the most significant predictors across all of the models, for Intrusion it showed $\beta = 0.62$, $p < 0.001$, for avoidance $\beta = 0.47$, $p < 0.001$ and for hyperarousal $\beta = 0.70$, $p < 0.001$. Similarly, Perceived Control also predicted hyperarousal, $\beta = 0.36$, $p = 0.1$, however it showed weaker results for both intrusion and avoidance. Out of the five personality traits, only agreeableness demonstrated a major result, where it negatively predicted hyperarousal, $\beta = -0.31$, $p = 0.4$, meanwhile the rest of the traits remained at a statistically non-significant level. The models displayed a moderate proportion of variance across its subscales, where R^2 ranged from 0.35 to 0.56, which reinforces that the predictors do relate meaningfully to the outcome.

Overall, the findings through the regression analysis were consistent with the studies shown by Budimir et al within Emotional Reactions to Cybersecurity Breach Situations: Scenario-Based Survey Study, who reported similar results where an individual's interpretation and their emotional appraisals of the incident were stronger predictors of a respondent's response than the Big Five personality traits. Both of the studies show an emotional reaction variable emerging as a dominant predictor, which supports a framework that recognizes the feeling of the individual and how it manifests their interpretation of the event, playing a central role in determining psychological outcomes. The study found Emotional Impact to be more emphasized across the model whereas personality traits showed minimal influence across all models. This may be a result of the usage of PCA derived components, where a broader capture of underlying dimensions provides a stronger predictive power.

Correlation Analysis

To continue the study of quantitative data, two-tailed Pearson correlation analyses that flagged significant correlations were conducted to examine the relationship between personality traits and both the total results from the IES-R scale and the PCA derived components that were gathered from section 4.2.3, Pearson's correlation coefficient (r) was used to determine the strength and direction of these linear relationships, significance was evaluated at the $p > 0.05$ level. These analyses aim to calculate the extent of which of the personality categories would be associated with the impact response to a cyberincident.

Between Conscientiousness and Perceived Control, $r = 0.275$, $p = 0.086$, we can pick up a positive correlation, indicating that participants who had a higher conscientiousness score tended to perceive themselves as more capable of managing a cyber incident. Although the relationship between the two did not reach statistical significance, it does provide a direction of association that aligns with theoretical expectations that link personality to preparedness.

The relationship between Emotional Stability and Emotional Impact on the other hand produced a negative correlation, $r = -0.264$, $p = 0.100$, pointing towards a suggestion that individuals with lower emotional stability experienced a greater impact and emotional response from cyberincidents. This trend is consistent with established psychological research where lower mental stability is associated with increased stress activity.

Now measuring the relationship between personality traits and a respondents' total score from the IES-R, Conscientiousness showcased a significant positive trend with the IES-R total ($r = 0.315$, $p = 0.048$), a finding that indicates that higher levels of conscientiousness were often tied to responses which had a greater reported impact following a cyber incident. Potentially, this reflects an increased sense of responsibility in the individual.

Similarly, the Imaginative Scaling showed a significant positive relationship with the IES-R total, $r = 0.315$, $p = 0.048$, implying that the respondents' that scored higher in the imaginative scale felt a stronger emotional impact from a cyberincident. This finding suggests that individuals which are more prone to reflect on their issues process cyber attacks to a greater degree.

Contrasting this, Emotional Stability displayed a negative yet non-significant relationship with the IES-R total, $r = -0.221$, $p = 0.170$, indicating that emotionally stable respondents experienced lower psychological distress.

Overall, the findings from the correlation analyses strongly imply that personality traits do play a measurable role in shaping an individual's response to a cyberincident. Other traits such as Extraversion, $r = -0.113$, $p = 0.487$ and Agreeableness, $r = -0.023$, $p = 0.89$, showed no major relationship with the IES-R total, so their results were not added to the analysis.

Scatter Graph Analysis

The Scatter graphs were generated through the use of SPSS and the simple scatter option. Once created, these graphs were then fitted with linear regression lines to visually display the direction and strength of the correlation shared between the Big Five personality traits and variables such as the IES-R total or the derivative variables from the PCA.

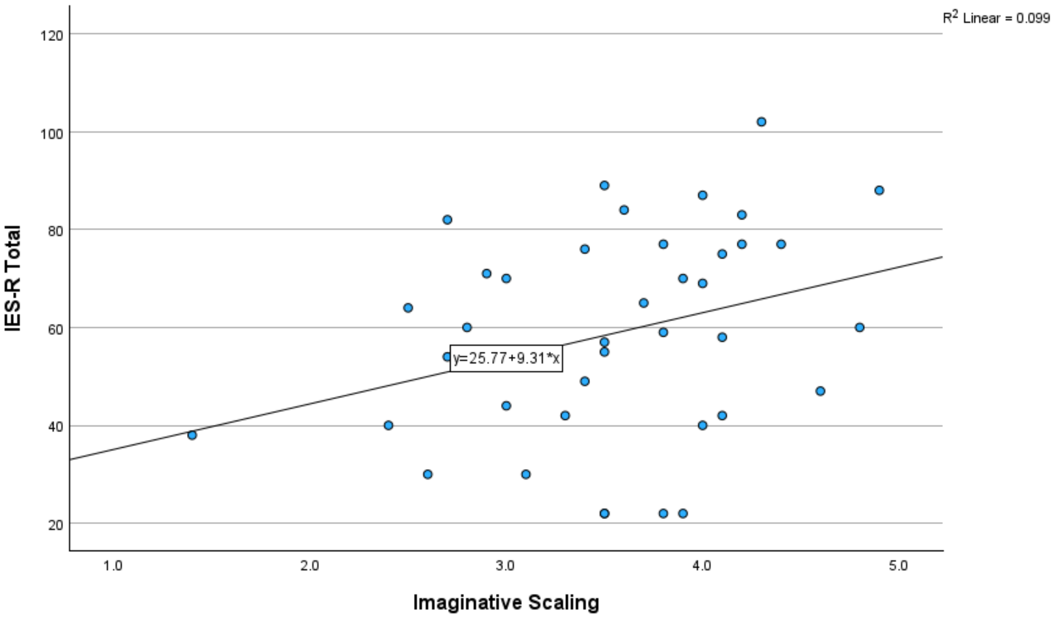

Figure 4.1: Imaginative Scaling and IES-R Total

Figure 4.1 illustrates the relationship between Imaginative Scaling and the IES-R Total score, where a positive relationship can be identified, y = 25.77 + 9.31x. Respondents who held a higher imaginative score within the survey reported greater psychological impact following the cyber incident, a finding that suggests that individuals with stronger imaginative tendencies are more prone to engage and reflect on the impact of the cyber attack they have experienced.

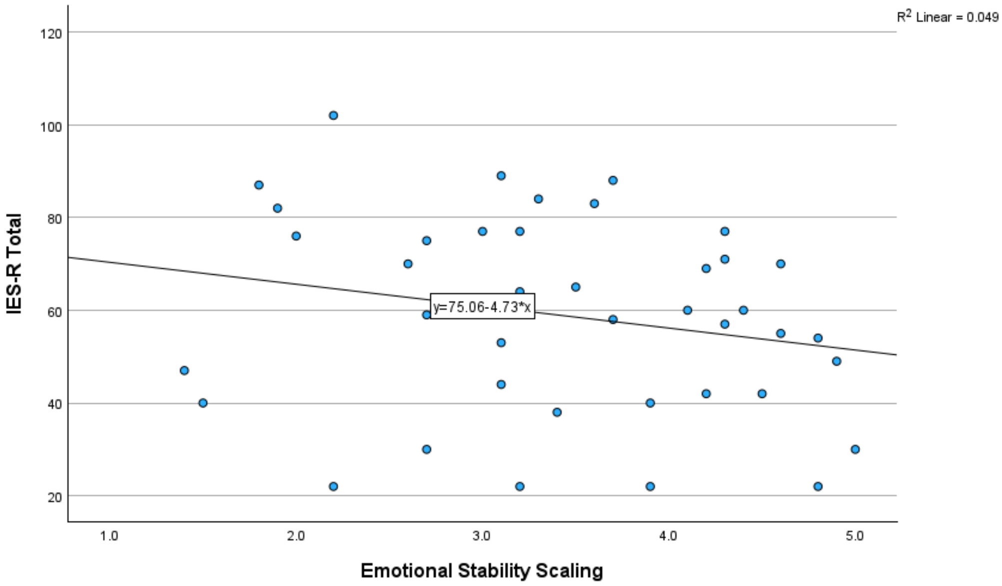

Figure 4.2: Emotional Stability and IES-R Total

In Figure 4.2 we can observe a stronger negative relationship between Emotional Stability and the IES-R Total scores, y = 75.06 - 4.73x. When respondents scored higher emotional stability, the scores regarding the impact of the cyberincident decreased, indicating that a stronger emotional stability functions to protect an individual's response to a cyberincident.

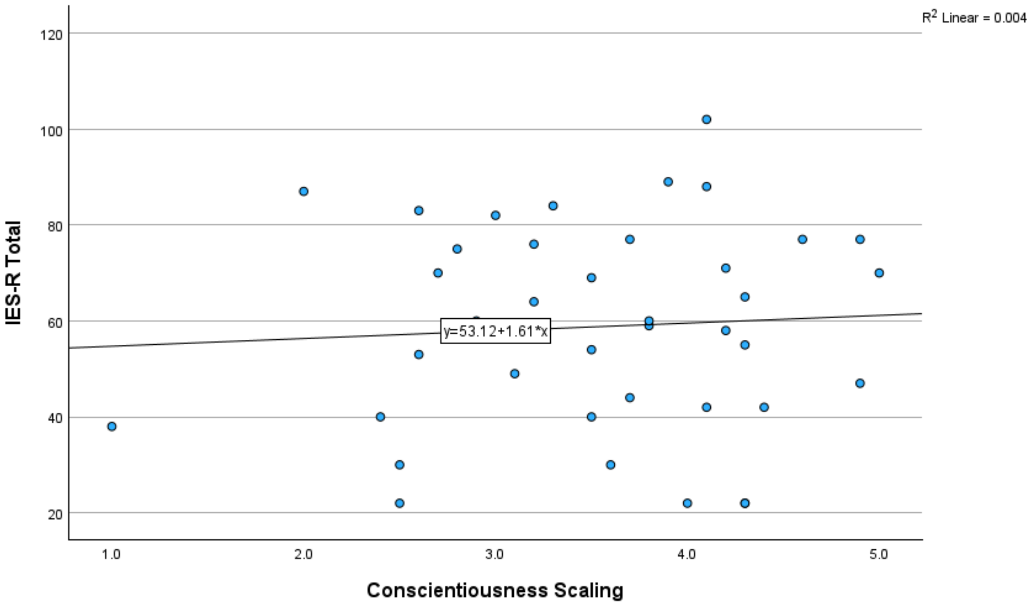

Figure 4.3: Conscientiousness and IES-R Total

Contrasting the previously analyzed figures, Figure 4.3 shows a small positive relationship between Conscientiousness and the IES-R total, y = 53.12 + 1.61x. While modest, the trend here suggests that a higher conscientiousness associates itself with a higher set of stress responses.

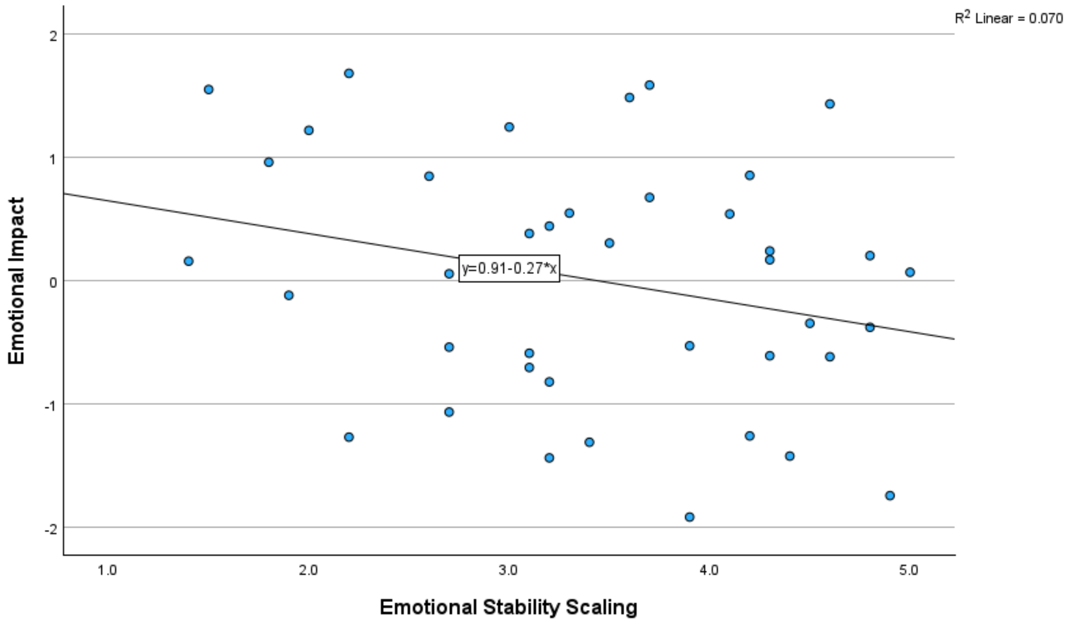

Figure 4.4: Emotional Stability and Emotional Impact

To further analyze the gathered data, scatter graphs were also composed from the components that were derived from the PCA. Figure 4.4 studies the relationship between Emotional Stability and Emotional Impact (y = 0.91-0.27x), where the two demonstrated a negative association, implying that individuals with a higher rating on emotional stability felt less of an emotional impact from the cyberincident and the aftermath of the incident.

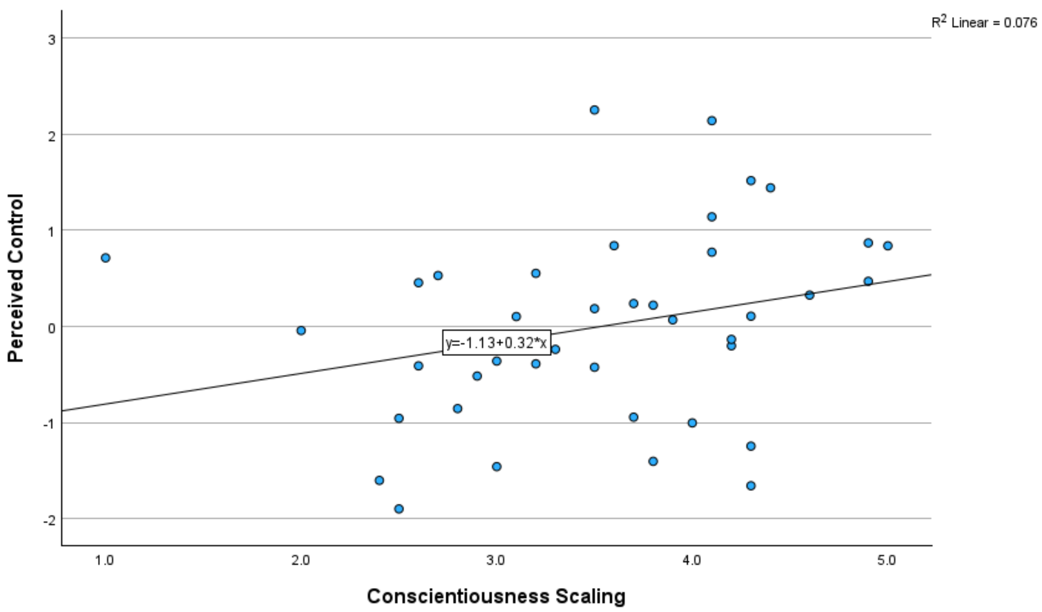

Figure 4.5: Conscientiousness Scaling and Perceived Control

Figure 4.5 analyses the relationship between Conscientiousness and perceived Control, revealing that there is a positive trend between the two items ($y = -1.13+0.32x$), evidencing that individuals with a higher conscientiousness had a higher sense of perceived control.

Overall, the scatter graphs reinforced previous findings from the correlation analysis through creating a visualization of the relationships between personality traits and the emotional responses. Agreeableness and Extraversion were traits that were not included in the IES-R total analysis due to showing no meaningful relationship with overall psychological impact.

4.3 Qualitative Findings

The qualitative analysis explored the respondents' experiences, how they reacted to the emotional impact from the cyber incident and picking out both the types of reactions where people differed or reacted similarly. Four themes were chosen after a full analysis of the data: Emotional Impact, Perceived Control, Within-Incident Experiences and Post-Incident Responses. The first two of these themes were taken from the PCA and overall these were chosen to provide contextual insight to the findings from the quantitative analysis.

Emotional Impact

Participants of the questionnaire frequently described the cyber incident they were a target of as emotionally impacting experiences, primarily significant ones too. These events often resulted in the respondent experiencing increased caution and awareness surrounding cybersecurity, many of the participants that felt this way also reported lingering emotional distress, such as a higher amount of distrust and heightened vigilance following a phishing attack. One participant noted that they had become " More paranoid about phishing and data destruction ", implying that their perception of online risk had been impacted to a point beyond formal awareness training. Similarly, another participant also responded with a similar statement, " The feeling of trust being exploited by a convincing phishing email has still stuck with me. " Further highlighting the lasting psychological effects from these incidents.

These responses indicate that the emotional reaction to an attack does go far beyond initial frustration, which contributes to the study's focus on whether an individuals' behaviour affects their response to an attack, in this case reflecting on behavioural learning and introspection from the respondent after the cyber incident. Several of the participants described how they used the experience as a motivation to improve their cybersecurity knowledge, showing that the emotional impact was used as motivation for adaptive change. This is a pattern that aligns with the quantitative findings that were derived from the IES-R scale, where many of the participants reported measurable psychological impact following the incidents.

Contrasting these results, not all respondents reported facing strong emotional distress. A small number primarily described feelings of frustration or annoyance rather than distress, evidencing the variation in emotional response across the participants' responses.

Perceived Control

Perceived Control was found to be a central factor when it came to shaping respondents' reactions of cyber incidents. Participants who felt that they were unable to influence the outcome of the attack often expressed frustration or distress, reflected in statements such as " The fact it was so easy and quick to lose control of what you always thought was safe. " and " The most impactful aspect was the realisation my actions contributed to the breach. The technical fallout was handled by the company, but the personal accountability and loss of confidence had a longer-lasting effect. " These responses further evidence the perceived preparedness of the individual or feelings of responsibility for their emotional response to the incident.

Other participants attributed responsibility to personal error, one of these participants noted that their personal experience of the situation was " Frustrating because it should have been obvious, " pointing towards self-evaluation following the incident. On the other hand, some respondents did not believe they were responsible for the attack, specifically when these respondents were targets of organisational data breaches. One such example, " A major identity data breach from a government department I had no choice but to give data to, resulting in successful identity theft costing significant effort to fix. " While also stating " I have never been meaningfully impacted by anything I had any opportunity to affect the "impact" of. "

Overall, participants who perceived themselves as having greater control frequently reported increased cyber awareness afterwards due to greater impact. These findings support the PCA-derived components of Perceived Control within the quantitative analysis, suggesting a consistency between statistical trends and lived experiences.

Within Incident Experiences

Respondents provided accounts describing how they felt during the incident, these experiences often contained immediate stress, confusion and distress. A pattern that was found is that phishing attacks were frequently associated with feelings of betrayal, especially in the case of a social engineering attack when the attacker would exploit the individuals' trust, an example of this, " The stress it gives at the moment and the stress that lingers on after. " Which illustrates both the immediate and continuous emotional strain.

Several of the participants also expressed surprise once finding out they had been the target of a cyberattack, with one specific respondent noting, " The fact a simple cooking blog was targeted by such an attack in the first place. " Others reflect on their initial awareness, often acknowledging their limited knowledge before the incident, " I probably wouldn't have installed the ransomware in the first place if I had known better. " These reflections demonstrate the uncertainty felt during the incident and how it influenced the emotional impact and then motivation to learn.

Despite these experiences, some of the participants again only reported mild annoyance within the incident, further implying variance in an individual's perceived severity or emotional engagement.

Post Incident Experiences

Post incident responses were primarily characterised by a large majority of the participants experiencing behavioural change alongside developing an increased level of cybersecurity awareness. A number of the respondents developed more cautious online behaviours, or began going out of their way to educate others. One respondent explained how " I try to help people and advise everyone to update software and double-check everything for guaranteed safety ", demonstrating the shift from a personal experience to taking action.

Other respondents experienced lasting effects after the incident beyond behavioural change. An example of this comes from one of the participants, " When the attack happened I felt hopeless and it left me scarred " While another participant reported, " More skeptical of people during work, trained to be better against cyberattacks " Alongside becoming better trained to deal with cyber attacks. Increased vigilance was also evident, with one of the respondents describing how, " I became a lot more obsessive about ensuring everything is secure. "

These findings suggest that the cyber incidents often produced long-term changes to the individual from the initial impact, reinforcing the broader pattern observed across the themes of the qualitative research where negative experiences contributed to higher awareness of cybersecurity.

4.4 Summary

The data analysis examined the relationship between the respondents' personality traits and their responses to the cyber incidents using both quantitative and qualitative data analysis. Descriptive statistics and reliability analysis proved that the study held an acceptable internal consistency across the measuring scales used for the study. The PCA identified two major components, Perceived Control and Emotional Impact, which were retained for further analysis.

Correlation Analysis revealed a statistically significant positive relationship between Conscientiousness and the IES-R total score, as well as the Imaginative Scaling and the IES-R total. Emotional Stability demonstrated a negative, albeit a non-significant, association with the scale. Further analysis identified a positive relationship between Conscientiousness and Perceived Control alongside a negative relationship between Emotional Stability and Emotional Impact. Agreeableness and Extraversion showed minimal correlation with the IES-R total.

The Multiple Regression Analysis produced alternative results, where the strongest and most consistent predictors for the IES-R results were the components derived from the PCA, Emotional Impact and Perceived Control, which in contrast the Big Five personality traits showed limited predictive value. This is a finding that supports Budimir et al's (2021) work in Emotional Reactions to Cybersecurity Breach Situations: Scenario-Based Survey Study, evidencing how an individual's interpretation of events play a more influential role in shaping the emotional response.

Qualitative analysis produced a total of four themes: Emotional Impact, Perceived Control, Within Incident Experiences and Post Incident Responses. Participants frequently described experiencing an emotional reaction as well as holding varying perceptions of responsibility, perceived control, distress experienced during incidents and a majority developed an increased sense of cybersecurity awareness following the cyberattacks. Many respondents reported behavioural changes and increased caution.

Overall, the analyses present a consistent set of patterns between the respondents linking personality measures as well as reported psychological and behavioural responses to cyber incidents.

Chapter 5: Discussion

5.1 Introduction

In this chapter the study's findings that were presented in chapter 4 will be discussed in relation to the study's aims of examining how individual personality traits influence the psychological response to the emotional impact from a cyber incident. By integrating quantitative results and the findings of the qualitative analysis, this chapter aims to interpret the observed relationships between personality traits, perceived control, emotional impact and the respondents' emotional responses. The discussion will analyze how the findings from the questionnaire address the research objectives and compare outcomes with existing research, evaluating the implications for cybersecurity awareness and individual preparedness.

5.2 Personality Traits and Emotional Impact

The quantitative findings of the study showed an association between personality traits and differing levels of psychological response amongst the respondents to cyber incidents. Conscientiousness and Imaginative Scaling showed a strong positive relationship with the IES-R total score, heavily implying that individuals with these traits often reported stronger emotional impact from a cyberattack. In contrast, Emotional Stability showed a negative correlation with the Emotional Impact, evidencing that the respondents who had higher emotional stability were not as affected by the cyberincident, although this association did not reach statistical significance as it was quite weak.

These findings suggest that an individual's traits relate to how they will process and interpret cyber incidents, but aren't the sole factor in determining their emotional response. Individuals with a higher score for Conscientiousness could be more attentive to the consequences or responsibility involved within a cyber incident, shifting their view of the impact. Similarly, individuals who scored a high Imaginative rating may view stressful incidents more cognitively, possibly amplifying the way they perceive the effects from it. Additionally, the absence from any strong relationships between the IES-R total and Agreeableness or Extraversion scaling within the study indicates that not all of the personality traits contribute equally to the response.

5.3 Perceived Control and Responses to Cyber Incidents

The PCA identified Perceived Control as one of the key dimensions underlying participant responses. Quantitative trends implied that a higher ranking of conscientiousness would link to individuals having a greater perception of preparedness and personal capability, while those who held a lower score for Perceived Control usually felt a greater level of frustration and concern over cyber incidents.

Qualitative findings went on to further reinforce this pattern as participants frequently described their frustration in response to the cyber incident, a number of respondents specifically wrote about feeling like they were unable to influence the outcome of the cyberincident or when responsibility was not in their hands. Some of the participants settled on the incidents being caused by personal error, meanwhile other respondents shifted responsibility when organisational failures were involved in the incident. The differing perceptions of control provide a highlight to the role of agency within an incident, alongside how it affects the emotional impact.

5.4 Emotional Impact and Behavioural Change

Both the quantitative and qualitative analyses showcased how cyber incidents can produce emotional reactions and how they varied depending on the personality traits of the individual targeted by the attack, emotions that were noted are anxiety, stress and distrust which were present in multiple of the respondents' accounts with varying degrees, a majority of these responses also frequently went beyond the incident, where some of the respondents wrote about experiencing lingering concerns or how the incident altered their perception of online safety as a whole.

Importantly as well, qualitative responses also showed that emotional impact would often be associated with behavioural adaptation. Many of the participants described that after the incident, they had become more cautious online, improving cybersecurity practices and some even going ahead to educate others following an incident. However, variability was still present across the participants, as some of the respondents didn't report any major change following the impact.

5.5 Experiences During and After the Incident

A consistent set of themes were also found regarding within incident and post incident experiences, revealing a progression in participant responses. During the incident, participants often described feeling confused, surprised or distressed, specific cyber attacks such as social engineering also caused feelings of betrayal within the individual. Following a majority of these cyber incidents, respondents described feelings of reflection as well as adopting behavioural change in response to the incident, often these changes took the form of increased vigilance and proactive security practices.

This progression shared across responses suggests that these cyber incidents have acted as learning experiences for the respondents, post incident reflection often influencing future behaviour. The findings here align with the broader objectives of the study, allowing for greater understanding on how individuals adapt following cybersecurity threats, or how variance was still present as some respondents did not think much of the cyber attack they were a target of.

5.6 Comparison with Existing Literature

The findings of this study broadly align with previous research that was examined within the literature review. Research by Budimir, Fontaine and Roesch (2021) also identified the negative emotional impact of cyber incidents, where both the study and their research found similar reactions such as frustration, anger and feelings of betrayal, all ranging in emotional intensity that varied in accordance to personality characteristics. Their work had also demonstrated that lower emotional stability was associated with stronger emotional reactions in cyber attack scenarios. Within the study, similar patterns were found, where participants frequently reported varied levels of distress, distrust and lingering emotions following cyber incidents. Furthermore, the Multiple Regression Analysis supported the results of Budimir et al's work, which evidenced that an individual's interpretation of events have a more influential role in shaping the emotional response than the Big Five personality traits.

The scenario based study by Budimir et al. (2021) further proposed that emotional reactions to cyber incidents consist of multiple dimensions, including emotional defense and proactive versus defensive responses. Within the study, this multidimensional structure was also reflected in the current findings, particularly through the emergence of the two components identified from the PCA, Emotional Impact and Perceived Control. Suggesting a comparable number of psychological processes underlying responses to cyber incidents.

The findings within the study also correspond with the UK Home Office (2025) report on ransomware victims, which found that individuals often experienced stress, guilt and behavioural changes following the attacks, with many of the participants becoming more cautious and aware of cybersecurity afterwards. Respondents of the present study reported the same increased awareness and behavioural adjustments post incident, reinforcing broader evidence of cyber incidents resulting in lasting psychological and behavioural consequences.

Overall, the study's findings were consistent with already existing literature, supporting the view that cybersecurity incidents both produce measurable emotional responses which are affected by the target's personality traits. These individuals then experience the incident from a perspective influenced by their personality, influencing post incident feelings.

5.7 Limitations

When interpreting these findings, several limitations from the study should be taken note of. While statistically significant, the sample size of respondents was still quite small, only reaching forty responses total from the questionnaire, this may limit the statistical reliability behind the study. As well as this, self reported data from the respondents could be influenced by a biased perspective or a subjective interpretation of experiences from the individual.

5.8 Summary

This chapter was used to interpret the study's findings, examining how personality traits relate to both the emotional response of a cyber incident and the impact felt by an individual due to these incidents. The results from the data gathered by the questionnaire suggest that personality traits, particularly Conscientiousness and Imaginative tendencies, are associated with the greatest variations in Emotional Impact and Perceived Control. Qualitative findings further pushed the view that cyber incidents commonly produce an emotional reaction from an individual, encouraging increased cybersecurity awareness and behavioural adaptation. Once put together, these findings indicate that individual personality traits do shape an individual's response to a cyber incident, more often than not shaping the way they will experience and respond to the emotional impact from the cyber attack.

Chapter 6: Project Management and Reflection

6.1 Aim & Objectives

The aims and objectives of this study evolved significantly during the research progress, reflecting a refinement of the intended scope for the study and research direction. Initially, the study sought to examine cyber threats in a broader manner with a focus on general market and internet safety, including understanding hacker behaviour and with a goal of contributing towards reducing cyber attacks. However, these initial objectives shifted and refined for the study to instead focus on investigating how individuals with differing personality traits emotionally respond to cyber threats and incidents, exploring the role of personality within cybersecurity while doing so.

These revised aims were successfully achieved through the use of a questionnaire which incorporated the IES-R, enabling the measurement of emotional impact following cyber incidents while also building off of previous academic works via utilizing an emotional scale that had not been incorporated within cybersecurity research before. Both quantitative and qualitative findings demonstrated patterns for Emotional Impact and Perceived Control, where most respondents also developed an increased level of cyber awareness following attacks, which directly addresses the study's objectives. This shift towards a psychological field with an individual focused perspective allowed for the study's research to provide meaningful insight into the human factors of cybersecurity.

6.2 Risks

Several of the risks that were identified throughout the project were primarily related to scope management, data collection and methodological challenges, the initial broad direction posed a risk of causing the study to produce findings that were not focused enough to write a dissertation with, refining the aims of the study towards personality and emotional response reduced this issue while also improving feasibility. Another one of the risks would be low participant engagement and potential difficulties analysing the questionnaire's responses, if the number of respondents was too low, there would have not been enough data to analyse and produce any findings, if there was no understanding on how to analyse the raw data, the findings also would not have been successfully produced. The study faced a risk akin to this one when initially it aimed to use Nvivo to process the qualitative data, however due to software failure ATLAS.ti was chosen in the end. These risks were mitigated through iterative planning, restructuring of objectives and the adapting of analysis methods to ensure that reliable results were obtained within the study's timeframe.

6.3 Project Management

Project management was carried out via the use of the MoSCoW planning framework, which was simple to adapt once the shift between the study's initial goals and the final ones after research. "Must Haves" elements included questionnaire development, ethical data collection and analysis that was carried out utilizing the IES-R alongside qualitative analysis. "Should Haves" tasks involved deeper comparative analysis with previous academic literature and the refinement of thematic coding for qualitative analysis. Broader cybersecurity implications were instead shifted onto "Could Haves" as the study's project focus narrowed. The use of MoSCoW enabled better prioritisation and more efficient time management even as the study shifted goals throughout its life cycle.

6.4 Reflection

Reflecting on the study, the refinement of the its aims and goals were definitely influenced due to the earlier objectives felt too broad to reasonably tackle within a dissertation and provide any reasonable findings, switching to a study that focused on human personality within cybersecurity that utilized previously unused emotional scales provided more academic contribution than the earlier goals. As well as this, the mixed-methods design of the questionnaire allowed for the research to both utilize quantitative findings using the IES-R scale and strengthen these through the research drawn from qualitative analysis. Utilizing software such as NVivo, ATLAS.ti and IBM SPSS also contributed to skill development within the use of quantitative research software and qualitative coding practices. Overall, the project lifecycle demonstrated how much research design can shift from the beginning to the end once further research is carried out.

Chapter 7: Conclusion

7.1 Points of Conclusion

Firstly, cyber incidents were shown to produce measurable emotional impact for many respondents, who frequently described suffering from distress alongside frustration or lingering distrust after being targeted. Qualitative findings also indicated that the emotional reaction of the individual extended beyond the immediate incident as many of the respondents described long term changes to their behaviour or how they interacted with the internet as a whole.

Quantitative analysis using a PCA identified two prominent dimensions influencing the emotional response from the respondents. Emotional Impact and Perceived Control. Quantitative analysis identified a trend where individuals who believed they were in control reported greater levels of emotional impact when targeted, often expressing frustration at the loss of control.

Phishing attacks were often reported to cause feelings of betrayal due to the exploitation of trust via social engineering. Many respondents initially underestimated the possible severity of the attack and the consequences, highlighting a gap between individual awareness and real time decision making. However, responses varied, as some of the participants didn't report an extreme emotional reaction, rather wording feelings of frustration with no lasting impact.

Post incident behaviour demonstrated clear adaptive outcomes. With most participants reporting gaining an increased sense of cyber awareness and education after the incident, in some cases even developing to educate others. Overall, the findings indicated that cyber incidents are not only technical events, but also personal experiences shaped by emotional impact and individual personality differences.

7.2 Recommendations

Based on the findings, a number of recommendations can be made to improve the emotional response to cyber incidents. Firstly, initiatives regarding awareness should address individuals' perception of control, seeing as how a higher sense of perceived control was often heavily associated with greater emotional impact to cyber incidents. Cybersecurity training would benefit from emphasizing that a number of cyber incidents are simply unavoidable, even with correct preparation. Reducing self blame to mitigate psychological distress.

Secondly, incorporating behavioural and psychological elements within cybersecurity training courses rather than solely focusing on technical prevention would lower the emotional impact from a cyber incident. Given that the emotional impact was the strongest predictor of distress within the Multiple Regression Analysis, training should aim to support emotional regulation and coping techniques following a cyber incident.

Additionally, establishing a work culture within corporations that prioritizes behavioural learning rather than finding a target to blame in response to cyber incidents would be beneficial, as the findings within the data analysis indicate that the respondents who felt responsible for an attack experienced a greater level of negative emotional distress, suggesting that supporting reporting environments would be one of the ways to reduce the emotional distress that follows after a cyber attack.

Finally, training and awareness strategies would benefit from being tailored to acknowledge user behaviour and personality traits rather than adopting an uniform approach. Aligning with the finding regarding how cognitive and emotional factors play a role in an individual's emotional response. Acknowledgment should be implemented within training aimed to prevent people that score highly on Emotional Stability, Conscientiousness and Imaginativeness of the Big Five scale, as those were the personality traits that shared the most correlation with overall emotional impact. Doing this would lower the cyber attacks emotional impact and encourage behavioural learning.

7.3 Further Research

While this study contributes statistical insight into emotional responses to cyber incidents, there is still further research that could be carried out for future investigations.

Future research could explore the same topic regarding personality traits and how an individual's cybersecurity knowledge and awareness would change depending on them while also using larger and more diverse samples to improve generalisability and statistical integrity. Expanding the respondent pool would allow the analysis of other aspects within the study, such as the respondent's culture, occupation or possibly age related differences within cyber incident experiences.

Additionally, future studies could also compare to this study via the use of different psychological instruments. The Big Five, while reliable, has been considered to be a dated method to measure personality, using another method to do so would provide further insight on how an individual reacts differently to cyber incidents due to their specific personality traits.

7.4 Conclusion

In conclusion, the study demonstrates that emotional impact to a cyber incident does differ between individuals due to their personality traits, often extending beyond the incident and leaving lingering feelings. In addition to this, an individual's sense of perceived control also went to significantly influence how participants interpreted and coped with the cyber incidents they were a target of, where many of the participants often reported becoming more cyber aware afterwards and going through behavioural adaptation following attacks. The study highlighted the importance of viewing cybersecurity beyond a solely technical lens, showcasing the importance of also viewing attacks through a human centered perspective for the sake of recognizing the emotional factors that will make up their response and their post incident feelings.

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Appendices

Appendix A: Project Initiation Document

Project Initiation Document

Project Details:

Project Title	*A Study on how Cyber Threats and Attacks Impact the General Market and Internet Safety*
Student	*Juan Rafael Moreno*
Course	*Cyber Security and Forensic Computing*
Project code	*PJS40*
Client	*N/A*
Supervisor	*David Williams*
Date	*September 2025*

1. **Client / Target Audience**

Guidance	My project would be aimed to inform a primary audience of cybersecurity analysts, practitioners and threat intel teams, as they would be the most likely to value the findings and benefit from my project. A possible secondary audience could be internet users that want to become more knowledgeable about possible cyber threats.

2. **Project background and problem to be solved**

<table>
<tr>
<td>Guidance</td>
<td>

The problem:

Threats posed by hackers and how it could affect unprepared businesses, for example somebody that doesn't know an example of what a modern phishing message could look like, or to be wary of digital cyber attacks or vulnerabilities in the way their programs are organized. An example of this kind of attack is reverse shell attacks.
Specifically talking about the Dark Web, organized attacks from hackers or the sharing of scripts would be the focus of my study project, but I may change this to be a project more focused on the research of cyber attacks.

Client:

A possible client for this project could be Law Enforcement & Cybercrime Units, as they would benefit from the insight into how the dark web commits fraud, illegal trades and endangers online safety

Topic:

There will be information around your chosen topic area which will help set the scene for your project so explain the context, such as statistics, existing solutions and why they’re not suitable, why is it important to achieve a solution, what is the benefit? For example if your project is for a manufacturing business, what is the global income generated by manufacturing? What is the UK income? Is the industry in decline or growing? Etc

The benefit and intent of my project is to increase awareness of possible threats for companies that intend to use the internet as a medium to communicate with their customers, as many of them use the Internet as a front.
Need to research why existing solutions are not suitable, perhaps because by the time this project will be written, it would have the most recent information available while other similar study projects would be outdated.

</td>
</tr>
</table>

3. **Project aim and objectives**

Guidance	What is the overall aim of the project? - To inform about the Deep and Dark Web and the dangers they could pose to companies What are the objectives that you hope to achieve in order to meet your aim? 1. To lower the percentage of attacks on businesses by any percentage, for example in the UK 43% of businesses have reported experiencing a cyber security breach or attack 2. Contextualize hackers as to explain their behaviours, for example, a hacker that took their scripts from somebody else may leave more of a digital footprint during their due to lack of experience 3. Attempt to research threats that are not widely documented, or attempt to discuss threats that are common but keep managing to pose a threat to companies [Phishing attacks make up 93% of cyber crimes against businesses] Cyber attacks Remember an objective is a goal, a desired outcome, something you aim to achieve, for example “increase turnover by 10%”. Often students just list tasks such as, “Carry out a literature review” but tasks usually say what you will do and how rather than why you will do something

4. **Project constraints**

Guidance	Any project will have external factors which will affect it. ● Deadline for the project is in May 2026 [32 weeks] ● The amount of time your supervisor is available - Unsure ● I can give time to the module but my other modules may pose some time constraints or take priority over the project at times ● Budget may become a constraint if I have to buy research articles or access to official scholarly articles ● The availability of specialist software to either simulate or inspect previous cyber attacks may limit the research I can do

5. **Project management**

Guidance	MoSCoW project management methodology
	
	Must Haves:
	
	• Cited writing with evidence of research done that provides reliable information and an answer to the project
	    • For example, I cannot deliver something that would not talk about the cyber threats posed by cyber attacks when the project is based around them
	
	Should Haves:
	
	• Simulations of how Malware or scripts would have been used to damage or steal a device, e.g simulate a payload attack on a computer then apply it to a hypothetical scenario within a company.
	• Evidence of the economy or companies either being affected negatively or positively by cyber attacks.
	    • Primarily negatively
	• Analysis of previous cyber attacks and how they function, would most likely go in tandem with the simulations
	
	• Surveys and Questionnaires to gather raw data of people's experiences with cyber attacks
	
	Could Haves:
	
	• Explanations on Phishing and Social Engineering attacks if they have any direct connection with the Deep Web or Dark Web
	• Bring up the CIA Triad (Confidentiality, Integrity and Accounting) and apply it to the attacks I will be investigating
	• Legal, ethical and professional implications of each attack, perhaps a post summary
	
	Won't Haves:
	
	• Information that is not relevant to the above
	
	Personal Classification Criteria:
	
	Must Haves would be consistently updated and worked on as the project is worked on, every bit of the work should apply to them. Should Haves be personal backing pieces of evidence that would require dedicated work to implement, these would take an allocated set of time. Could Haves are things that could be added onto the project if needed and I see them lending me a higher mark, Won't Haves is information that I will not actively seek out to research unless necessary.

1 week 1 to 1, half an hour. Come in with a clear plan on what I want to do. Group meeting for an hour. Next hour I will drop in with a group. If I need stuff on top of this, make appointments.

6. Tasks and timescales

Guidance	• To determine tasks and timescales you need to plan everything: • The timeline is approx 32 weeks • Break up those 32 weeks into large 'chunks' and allocate a project stage/major task to each one • The stages of your project will depend on the choices you have made for: **I want to discuss with with David Williams my Supervisor, I haven't had a chance to yet and I apologize for this** This is a **study** project that will be using the MoSCoW project metho Once your Gantt Chart is completed and added as an appendix, here you put in a table listing **the main stages and dates only**.

7. **Facilities and resources**

Guidance	All projects will need resources (equipment and people) in order to be completed. List:
	
	● Computers / IT to carry out the research:
	    ● I will most likely use either my own pc and my laptop for this
	
	● Software required to carry out and write my project in:
	    ● Tools to write and plan with, Microsoft Office or Google Workspace
	    ● Web browsers to research my project
	    ● Perhaps specific tools to simulate the attacks to show how they work
	        ● Programming Software
	        ● Virtual Machines
	        ● Script reading software
	    ● If needed, specialist software to research the deep web
	
	Make sure you mention any constraints on their availability.

8. **Project risks**

Guidance	*All* projects have risks and yours is no exception. We need to know what risks you think might happen and how you might mitigate them (reduce the chance of them happening). Fill out the table below with the **most likely** risks (do a search for the top project risks) to **your** project? - There is a risk that the project may come off as plagiarized if I rely too much on references and citations

No	**Description**	**Likelihood (high, medium, low)**	**Impact**	**Mitigation/Avoidance**
Example	*Laptop failure*	*Medium*	*Cause delays to the project timeline and data loss*	*Back up work to a USB* ***and** another source*
1	*Researching about the Deep Web and Dark Web is fine from an outside perspective, but I am worried about the legality and risk of if I would have to specifically enter said areas*	*Medium*	*May lead to the project being reworked.*	*Research and make sure that I do not break the law while working on the project. Access such things under legal grounds.*
2	*Computer failure*	*Low*	*Cause delays to project timeline and data loss*	*Back up work consistently, work on laptop if anything happens.*
3	*Companies that I research on may desire to not have their previous breaches looked into*	*Low*	*Will have to research other attacks instead*	*Make sure that whichever data breach I'm discussing has been well documented*

9. **Project deliverables**

Guidance	*All* projects produce 'things' which need to be listed here
	
	For a study or research project:
	● What documents and artefacts will be produced? E.g. primary research tool, datasets, data analysis and your report?
	
	● I plan to provide a primary document with the majority of the writing within it and links to the artefacts
	● Artefacts produced would be either:
	    ● Graphs displaying the effects of cyber attacks
	    ● Scripts written to simulate a specific kind of cyber attack

10. **Research**

Guidance	What secondary research do you need to do? What are the starting points for your research? (e.g. specific books or papers, Google Scholar, Discovery) and what primary research do you plan to do (if any)? Primary research may include attempting to interview companies on the cyber attacks they've experienced, or create surveys for the average person to describe their experiences with them. Secondary research would be taken from scholarly articles from Google Scholar, Discovery or books that are relevant to the topic and are expertly reviewed. Academic papers. Specific Names: University Library has the Search Library Resources Google Scholar Scopus Credo Reference

11. Legal, ethical, professional, social issues

Guidance	What are the legal/ethical/professional/social issues that may impose constraints on the project? How will you ensure that they will be addressed, or what steps will you take to avoid/mitigate their effects? What laws must you comply with? Whatever project work you are doing, you must consider its security implications, for the data you generate or use, or for the software artefact itself. Please describe how you are taking these into account. There is also a question about security on the ethics review form

Legal:

- Legality of this mostly goes around the fact that accessing the deep web and dark web is illegal unless under specific circumstances

Ethical:

- Ethically it is important to warn and inform businesses on possible threats as a cyber security attack may economically cripple the company permanently following a loss of reputation after a data breach, or to ensure that the customers can trust the company to keep their private information safe.
- Then there is also the argument between an individual's privacy and anonymity against the need to control illegal activities.

Professional:

- Cybersecurity focused, can also apply to law enforcement and digital privacy fields.

12. Supervisor meetings

Guidance	I have not managed to arrange a consistent schedule of meetings with my advisor yet, I am trying to look into arranging this

13. Declarations

Please tick the following declarations:

Usage (optional)	
X	I give permission for this document to be made available to other students as examples of previous work.
Authenticity of work	
✓	I confirm that I have read the University rules in respect of plagiarism and student misconduct.
✓	I understand that if I use work from an external source, I must reference and cite the source in any work that I produce.
Ethics	
✓	I understand that ethical approval will be needed for this project regardless of whether I carry out primary research or not.
✓	I understand that if I do conduct primary research, I must include all raw data and research documentation in my final project report.
✓	I understand that not including the raw data can mean potentially failing the project.
✓	I understand that not obtaining ethical approval is grounds for academic misconduct and possible project failure.
Use of AI Artificial Intelligence (AI), such as ChatGPT, can be used as a tool to help assist and inform the initial development of your work but it is not a replacement for your own critical thinking and analysis. We require that your work is your own original content, demonstrating your knowledge, skills, and critical thinking abilities. To that end, AI-generated content must not be included in <u>any work</u> you submit for assessment unless suitably referenced. Not doing so is a breach of the University's Academic Regulations as outlined in the Student Conduct Policy and constitutes an assessment offence.	
✓	I have read the University AI policy and if relevant, I will acknowledge the use of AI in any work I produce for this project with appropriate referencing.
✓	I understand that not appropriately referencing the use of AI is an assessment offence equal to plagiarism and may result in a penalty which remains on my student record.

Name: Juan Rafael Moreno
Date: 30/09/2025

Appendix B : Ethics Certificate

UNIVERSITY OF PORTSMOUTH

Certificate of Ethics Review

Project title: A study on how individuals of differing personality types are impacted by cyber attacks

Name:	Juan Rafael Moreno	**User ID**:	up2196 0 59	**Application date**:	27/02/2026 16:19:16	**ER Number**:	TETHIC-2026-113126

You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are **Elisavet Andrikopoulou, Kirsten Smith**

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

2 University Policy
3 Safety on Geological Fieldwork

It is also your responsibility to follow University guidance on Data Protection Policy:

4 General guidance for all data protection issues
5 University Data Protection Policy

Which school/department do you belong to?: **School of Computing**
What is your primary role at the University?: **Undergraduate Student**
What is the name of the member of staff who is responsible for supervising your project?: **David Williams**
Is the study likely to involve human subjects (observation) or participants?: Yes
Will you gather data about people (e.g. socio-economic, clinical, psychological, biological)?: No
Will your data collection be strictly limited to gathering anonymous insights about a particular artefact or research question (e.g. opinions, feedback)?: Yes
Confirm whether and explain how you will use participant information sheets and apply informed consent.: The participant population sheet will be available online, informed consent will be captured by using the template and linking into the first page of the Microsoft form.
Confirm whether and explain how you will maintain participant anonymity and confidentiality of data collected: No PII will be captured, only experiences and opinions regarding aspects of Cyber Security.
Will the study involve National Health Service patients or staff?: No
Do human participants/subjects take part in studies without their knowledge/consent at the time, or will deception of any sort be involved? (e.g. covert observation of people, especially if in a non-public place): No
Will you collect or analyse personally identifiable information about anyone or monitor their communications or on-line activities without their explicit consent?: No
Does the study involve participants who are unable to give informed consent or are in a dependent position (e.g. children, people with learning disabilities, unconscious patients, Portsmouth University students)?: No
Are drugs, placebos or other substances (e.g. food substances, vitamins) to be administered to the study participants?: No
Will blood or tissue samples be obtained from participants?: No
Is pain or more than mild discomfort likely to result from the study?: No
Could the study induce psychological stress or anxiety in participants or third parties?: No Will the study involve prolonged or repetitive testing?: No
Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No
Are there risks of significant damage to physical and/or ecological environmental features?: No
Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No
Does the project involve animals in any way?: No
Could the research outputs potentially be harmful to third parties?: No Could your research/artefact be adapted and be misused?: No

Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: Yes
Please briefly explain any security implications that may arise from the project/deliverables.: No security implications will arise, but as a cyber project that is looking how Malware affects Internet safety, it would be of interest to security organizations by its nature. The project will not be identifying new threats or vulnerabilities that aren't already well known in the public domain.
Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.
Supervisor comments:

Supervisor's Digital Signature: **david.williams2@port.ac.uk**

Date: **27/02/2026**

Faculty Ethics Committee Review

Ethics Rep comments:

Faculty Ethics Committee Member's Digital Signature(s): **elisavet.andrikopoulou@port.ac.uk**

Date: **27/02/2026**

Appendix C : Artefact and Consent Form

Consent Form was embedded into the questionnaire's first page

A study on how individuals of differing personality types are impacted by cyber attacks

Name and Contact Details of Researcher: Juan Rafael Moreno, up2196059@myport.ac.uk

Contact Details of Supervisor: David Williams, david.williams2@port.ac.uk

University Data Protection Officer: Samantha Hill, 023 9284 3642 or information-matters@port.ac.uk

Ethics Committee Reference Number: TETHIC-2026-113126

This study is concerned with cyber attacks and how people of differing personality types may experience different impacts. We are seeking participants who should have experienced a cyber attack personally or professionally. Participation in the research would require you to complete the attached digital questionnaire and take approximately 15 minutes of your time.

Please read and understand the provided information sheet, Ethics Document.docx.

* Indicates required question

1.

I confirm that I have read and understood the information sheet ([Ethics Document.docx]) dated February 2026 (version 1.1) for the above study. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily. *

Check all that apply.

□ I agree

2.

I understand that my participation is voluntary and that I am free to up until the point that I submit a response to the questionnaire. Thereafter, my information is not personally identifiable and cannot be withdrawn from the dataset. *

Check all that apply.

□ I agree

3.

I understand that my participation is voluntary and that I am free to withdraw at any time (or add the date after which withdrawal will not be possible) without giving any reason. *

Check all that apply.

□ I agree

4.

□ I agree to partake in the study *

Check all that apply.

□ I agree

5.

Questions regarding cyber attacks

These questions are focused on the cyber attack that affected you.

Some of these questions will be ranged from 1 to 5, which are as follows:

1 - Strongly Disagree

2 - Disagree

3 - Undecided

4 - Agree

5 - Strongly Agree

Questions regarding an individual's experiences with cyber attacks *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ Would you agree you have been a target of frequent Cyber Attack?

⬭ ⬭ ⬭ ⬭ ⬭ Would you say you were impacted by the attack?

⬭ ⬭ ⬭ ⬭ ⬭ Did your personality change after the attack?

⬭ ⬭ ⬭ ⬭ ⬭ Do you feel if you had been trained to deal with cyber attacks, the impact would have been minimized?

⬭ ⬭ ⬭ ⬭ ⬭ Would you say at the time you felt secure against the threat of a cyber attack?

⬭ ⬭ ⬭ ⬭ ⬭ Do you believe it happened because you did something wrong?

⬭ ⬭ ⬭ ⬭ ⬭ Did you feel your trust was betrayed?

⬭ ⬭ ⬭ ⬭ ⬭ Were you worried similar situations may happen in the future?

⬭ ⬭ ⬭ ⬭ ⬭ Did you feel you could do anything about the situation?

⬭ ⬭ ⬭ ⬭ ⬭ Were you afraid your security and the security of people close to you could be jeopardized?

6.

What kind of cyber attacks have you been a target of? *

Check all that apply.

□ Malware Attack

□ Denial-of-Service (DoS) Attack

□ Phishing

□ Spoofing

□ Identity-Based Attacks

□ Code Injection Attacks

□ Social Engineering Attack

□ Insider Threats

□ DNS Tunelling

□ Other: ____

7.

Which cyber attack would you say impacted you the most? *

Mark only one oval.

⬭ Malware Attack

⬭ Denial-of-Service (DoS) Attack

⬭ Phishing

⬭ Spoofing

⬭ Identity-Based Attacks

⬭ Code Injection Attacks

⬭ Social Engineering Attack

⬭ Insider Threats

⬭ DNS Tunelling

⬭ Other:

8.

How would you describe the initial impact of a cyber attack?

9.

If you faced different kinds of cyber attacks, did the impact change between the attacks?

10.

Would you say that if you had been better trained to handle cyber attacks, you could have lowered the impact? Why?

This section focuses on behaviour and the emotional impact of the attack

Questions taken from IPIP Big-Five Factors and Impacts of Events Scaled – Revised.

Some of these questions will be ranged from 1 to 5, which are as follows:

1 - Strongly Disagree

2 - Disagree

3 - Undecided

4 - Agree

5 - Strongly Agree

11.

Extraversion Scaling *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ I am the life of the party

⬭ ⬭ ⬭ ⬭ ⬭ I feel comfortable around people

⬭ ⬭ ⬭ ⬭ ⬭ I start conversations

⬭ ⬭ ⬭ ⬭ ⬭ I talk to a lot of different people at parties

⬭ ⬭ ⬭ ⬭ ⬭ I don't mind being the center of attention

⬭ ⬭ ⬭ ⬭ ⬭ I don't talk a lot

⬭ ⬭ ⬭ ⬭ ⬭ I keep in the background

⬭ ⬭ ⬭ ⬭ ⬭ I have little to say

⬭ ⬭ ⬭ ⬭ ⬭ I don't like drawing attention to myself

⬭ ⬭ ⬭ ⬭ ⬭ I am quiet around strangers

12.

Agreeableness Scaling *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ I am interested in people

⬭ ⬭ ⬭ ⬭ ⬭ I sympathize with others' feelings

⬭ ⬭ ⬭ ⬭ ⬭ I have a soft heart

⬭ ⬭ ⬭ ⬭ ⬭ I take time out for others

⬭ ⬭ ⬭ ⬭ ⬭ I feel others' emotions

⬭ ⬭ ⬭ ⬭ ⬭ I make people feel at ease

⬭ ⬭ ⬭ ⬭ ⬭ I am not really interested in others

⬭ ⬭ ⬭ ⬭ ⬭ I insult people

⬭ ⬭ ⬭ ⬭ ⬭ I am not interested in other people's problems

⬭ ⬭ ⬭ ⬭ ⬭ I feel little concern for others

13.

Conscientiousness Scaling *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ I am always prepared

⬭ ⬭ ⬭ ⬭ ⬭ I pay attention to details

⬭ ⬭ ⬭ ⬭ ⬭ I get chores done right away

⬭ ⬭ ⬭ ⬭ ⬭ I like order

⬭ ⬭ ⬭ ⬭ ⬭ I follow a schedule

⬭ ⬭ ⬭ ⬭ ⬭ I am exacting in my work

⬭ ⬭ ⬭ ⬭ ⬭ I leave my belongings around

⬭ ⬭ ⬭ ⬭ ⬭ I make a mess of things

⬭ ⬭ ⬭ ⬭ ⬭ I often forget to put things back in their proper place

⬭ ⬭ ⬭ ⬭ ⬭ I shirk my duties

14.

Emotional Stability Scaling *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ I am relaxed most of the time

⬭ ⬭ ⬭ ⬭ ⬭ I seldom feel blue

⬭ ⬭ ⬭ ⬭ ⬭ I get stressed out easily

⬭ ⬭ ⬭ ⬭ ⬭ I worry about things

⬭ ⬭ ⬭ ⬭ ⬭ I am easily disturbed

⬭ ⬭ ⬭ ⬭ ⬭ I get upset easily

⬭ ⬭ ⬭ ⬭ ⬭ I change my mood a lot

⬭ ⬭ ⬭ ⬭ ⬭ I have frequent mood swings

⬭ ⬭ ⬭ ⬭ ⬭ I get irritated easily

⬭ ⬭ ⬭ ⬭ ⬭ I often feel blue

15.

Imaginative Scaling *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ I have a rich vocabulary

⬭ ⬭ ⬭ ⬭ ⬭ I have a vivid imagination

⬭ ⬭ ⬭ ⬭ ⬭ I have excellent ideas

⬭ ⬭ ⬭ ⬭ ⬭ I am quick to understand things

⬭ ⬭ ⬭ ⬭ ⬭ I use difficult words

⬭ ⬭ ⬭ ⬭ ⬭ I spend time reflecting on things

⬭ ⬭ ⬭ ⬭ ⬭ I am full of ideas

⬭ ⬭ ⬭ ⬭ ⬭ I have difficulty understanding abstract ideas

⬭ ⬭ ⬭ ⬭ ⬭ I am not interested in abstract ideas

⬭ ⬭ ⬭ ⬭ ⬭ I do not have a good imagination

Questions regarding the emotional impact of the cyber attack *

Mark only one oval per row.

1 2 3 4 5

⬭ ⬭ ⬭ ⬭ ⬭ Any reminder brought back feelings about it?

⬭ ⬭ ⬭ ⬭ ⬭ I had trouble staying asleep

⬭ ⬭ ⬭ ⬭ ⬭ Other things keep making me think about it

⬭ ⬭ ⬭ ⬭ ⬭ I felt irritable and angry

⬭ ⬭ ⬭ ⬭ ⬭ I avoided letting myself get upset when I thought about or was reminded of it

⬭ ⬭ ⬭ ⬭ ⬭ I thought about it when I didn't mean to

⬭ ⬭ ⬭ ⬭ ⬭ I felt as if it hadn't happened or wasn't real

⬭ ⬭ ⬭ ⬭ ⬭ I stayed away from reminders about it

⬭ ⬭ ⬭ ⬭ ⬭ Pictures about it popped into my mind

⬭ ⬭ ⬭ ⬭ ⬭ I was jumpy and easily startled

⬭ ⬭ ⬭ ⬭ ⬭ I tried not to think about it

⬭ ⬭ ⬭ ⬭ ⬭ I was aware that I still had a lot of feelings about it, but I didn't deal with them

⬭ ⬭ ⬭ ⬭ ⬭ My feelings about it were kind of numb

⬭ ⬭ ⬭ ⬭ ⬭ I found myself feeling or acting like I was back at that time

⬭ ⬭ ⬭ ⬭ ⬭ I had trouble falling asleep

⬭ ⬭ ⬭ ⬭ ⬭ I had waves of strong feelings about it

⬭ ⬭ ⬭ ⬭ ⬭ I tried to remove it from my memory

⬭ ⬭ ⬭ ⬭ ⬭ I had trouble concentrating

⬭ ⬭ ⬭ ⬭ ⬭ Reminders of it caused me to have physical reactions

⬭ ⬭ ⬭ ⬭ ⬭ I had dreams about it

⬭ ⬭ ⬭ ⬭ ⬭ I felt watchful or on-guard

⬭ ⬭ ⬭ ⬭ ⬭ I tried not to talk about it

⬭ ⬭ ⬭ ⬭ ⬭ Any reminder brought back feelings about it?

⬭ ⬭ ⬭ ⬭ ⬭ I had trouble staying asleep

⬭ ⬭ ⬭ ⬭ ⬭ Other things keep making me think about it

⬭ ⬭ ⬭ ⬭ ⬭ I felt irritable and angry

⬭ ⬭ ⬭ ⬭ ⬭ I avoided letting myself get upset when I thought about or was reminded of it

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⬭ ⬭ ⬭ ⬭ ⬭ I stayed away from reminders about it

⬭ ⬭ ⬭ ⬭ ⬭ Pictures about it popped into my mind

⬭ ⬭ ⬭ ⬭ ⬭ I was jumpy and easily startled

⬭ ⬭ ⬭ ⬭ ⬭ I tried not to think about it

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⬭ ⬭ ⬭ ⬭ ⬭ I found myself feeling or acting like I was back at that time

⬭ ⬭ ⬭ ⬭ ⬭ I had trouble falling asleep

⬭ ⬭ ⬭ ⬭ ⬭ I had waves of strong feelings about it

⬭ ⬭ ⬭ ⬭ ⬭ I tried to remove it from my memory

⬭ ⬭ ⬭ ⬭ ⬭ I had trouble concentrating

⬭ ⬭ ⬭ ⬭ ⬭ Reminders of it caused me to have physical reactions

⬭ ⬭ ⬭ ⬭ ⬭ I had dreams about it

⬭ ⬭ ⬭ ⬭ ⬭ I felt watchful or on-guard

⬭ ⬭ ⬭ ⬭ ⬭ I tried not to talk about it

17.

Would you say your personality was impacted by the cyber attack(s)? Please elaborate.

18.

Which facet of the cyber attack would you say impacted you the most? Please explain.

Appendix D : Raw Data Collected

[Raw Data Collected]

Appendix E : Correlation Analysis Matrix

Correlations

		Extraversion Scaling	Agreeableness Scaling	Conscientiousness Scaling	Emotional Stability Scaling	Imaginative Scaling	IES-R Intrusion	IES-R Avoidance	IES-R Hyperarousal	IES-R Total	Emotional Impact	Control and Preparedness
Extraversion Scaling	Pearson Correlation	1	.402*	-.150	.331*	-.112	-.075	-.180	-.061	-.113	.005	.107
	Sig. (2-tailed)		.010	.355	.037	.490	.645	.267	.711	.487	.977	.512
	N	40	40	40	40	40	40	40	40	40	40	40
Agreeableness Scaling	Pearson Correlation	.402*	1	.220	.068	.223	.082	-.126	-.033	-.023	.314*	.181
	Sig. (2-tailed)	.010		.172	.677	.167	.615	.440	.841	.890	.049	.264
	N	40	40	40	40	40	40	40	40	40	40	40
Conscientiousness Scaling	Pearson Correlation	-.150	.220	1	.328*	.387*	.050	.091	.041	.065	.191	.275
	Sig. (2-tailed)	.355	.172		.039	.014	.758	.575	.802	.689	.237	.086
	N	40	40	40	40	40	40	40	40	40	40	40
Emotional Stability Scaling	Pearson Correlation	.331*	.068	.328*	1	-.230	-.262	-.188	-.163	-.221	-.264	.345*
	Sig. (2-tailed)	.037	.677	.039		.153	.103	.244	.315	.170	.100	.029
	N	40	40	40	40	40	40	40	40	40	40	40
Imaginative Scaling	Pearson Correlation	-.112	.223	.387*	-.230	1	.362*	.240	.281	.315*	.442**	-.024
	Sig. (2-tailed)	.490	.167	.014	.153		.022	.136	.079	.048	.004	.885
	N	40	40	40	40	40	40	40	40	40	40	40
IES-R Intrusion	Pearson Correlation	-.075	.082	.050	-.262	.362*	1	.817**	.905**	.965**	.625**	.178
	Sig. (2-tailed)	.645	.615	.758	.103	.022		<.001	<.001	<.001	<.001	.272
	N	40	40	40	40	40	40	40	40	40	40	40
IES-R Avoidance	Pearson Correlation	-.180	-.126	.091	-.188	.240	.817**	1	.780**	.921**	.436**	.188
	Sig. (2-tailed)	.267	.440	.575	.244	.136	<.001		<.001	<.001	.005	.246
	N	40	40	40	40	40	40	40	40	40	40	40
IES-R Hyperarousal	Pearson Correlation	-.061	-.033	.041	-.163	.281	.905**	.780**	1	.942**	.621**	.259
	Sig. (2-tailed)	.711	.841	.802	.315	.079	<.001	<.001		<.001	<.001	.107
	N	40	40	40	40	40	40	40	40	40	40	40
IES-R Total	Pearson Correlation	-.113	-.023	.065	-.221	.315*	.965**	.921**	.942**	1	.593**	.216
	Sig. (2-tailed)	.487	.890	.689	.170	.048	<.001	<.001	<.001		<.001	.180
	N	40	40	40	40	40	40	40	40	40	40	40
Emotional Impact	Pearson Correlation	.005	.314*	.191	-.264	.442**	.625**	.436**	.621**	.593**	1	.000
	Sig. (2-tailed)	.977	.049	.237	.100	.004	<.001	.005	<.001	<.001		1.000
	N	40	40	40	40	40	40	40	40	40	40	40
Control and Preparedness	Pearson Correlation	.107	.181	.275	.345*	-.024	.178	.188	.259	.216	.000	1
	Sig. (2-tailed)	.512	.264	.086	.029	.885	.272	.246	.107	.180	1.000	
	N	40	40	40	40	40	40	40	40	40	40	40

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Appendix F : Participant Information Sheet

Ethics Document.docx

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PARTICIPANT INFORMATION SHEET

Title of Project: A Study on how Individuals of Differing Personality Types Are Impacted by a Cyber Attack

Name and Contact Details of Researcher(s):

Juan Rafael Moreno – up2196059@myport.ac.uk

Name and Contact Details of Supervisor:

David Williams – david.williams2@port.ac.uk

Ethics Committee Reference Number: TETHIC-2026-113126

1. Invitation

I would like to invite you to take part in my research study. Joining the study is entirely up to you, before you decide I would like you to understand why the research is being done and what it would involve for you. I would suggest this should take about 15 minutes. Please feel free to talk to others about the study if you wish. Do ask if anything is unclear.

I am Juan Rafael Moreno, a third year student at the University of Portsmouth who is creating this questionnaire to gather information for my Final Year Project.

2. Study Summary

This study is concerned with cyber attacks and how people's behaviour may be affected by them. We are seeking participants who should have experienced a cyber attack personally or professionally. Participation in the research would require you to complete the attached digital questionnaire and take approximately 15 minutes of your time.

3. What is the purpose of the study?

The purpose of this study is to gather first-hand experience of the effect of cyberattacks on the victims, it is primarily an educational project meant to inform and warn others on the possible dangers that one can stumble on within the Internet. - Illustrating the severity of malware attacks. Output of analysis, SPSS state.

4. Why have I been invited?

You were chosen as an individual that has prior experiences with the usage of the Internet. I am inviting you to take this questionnaire regardless of whether you have had any experience with being the target of a cyber attack, as either set of experience would prove helpful to my project.

5. Do I have to take part?

No, taking part in this research is entirely voluntary. It is up to you to decide if you want to volunteer for the study. We will describe the study in this information sheet. If you agree to take part, we will then ask you to sign the attached consent form, dated February 2026, version number, **1.1.**

6. What will happen to me if I take part?

If you are taking part in the questionnaire, then you will fill out the proposed questions within the digital document over a timespan of approximately 15 minutes, your data will be stored safely once done and that will be all that you have to do to take part. The information inputted into the questionnaire will be used within my Final Year Project.

7. Expenses and payments

No payments will be needed.

8. Anything else I will have to do?

There are no extra steps that you will have to do.

9. What data will be collected and / or measurements taken?

The data collected will be personal experiences with cyber attacks, specifications of which cyber attacks and the impact that the attack has had on your life. Additional information will be filling out a secondary section, where you will rate your personality based on questions from the Big Five Personality Model and other similar models.

10. What are the possible disadvantages, burdens and risks of taking part?

There are no major disadvantages to this questionnaire other than having to manually fill it out with your own personal experiences, however a risk is that the information within the answers that are sent to me could accidentally be leaked if a bad job is done ensuring their safety.

11. What are the possible advantages or benefits of taking part?

You will not receive any direct personal benefits from participating, however the results of this questionnaire will substantially support my Final Year Project.

12. Will my data be kept confidential?

The raw data and any data that identifies an individual will be kept secured and confidential, however the responses to the questionnaire could possibly be cited in the work itself.

How will we use information about you?

The department of Computing, Mathematics and Physics of the University of Portsmouth wishes to process your personal data (that is, collect, use, store and destroy data that identifies you) as part of a survey to analyse how cybercrime impacts a user's behaviour. If you have any queries about this project please contact Juan Rafael Moreno at up2196059@myport.ac.uk or if you have any general queries about how your data will be processed, please contact the University's Data Protection Officer, Samantha Hill, using any of the following contact details:

Samantha Hill, 023 9284 3642 or data-protection@port.ac.uk, Mercantile House, Hampshire Terrace, Portsmouth, PO1 2EG

The department of Computing, Mathematics and Physics legal basis for processing your personal data is that the processing is necessary for the performance of a task carried out in the public interest – that is, scientific or historical research purposes. If you withdraw from this research study we may still rely on this lawful basis to continue using your data if your withdrawal would be of significant detriment to the research study aims. We will always have in place appropriate safeguards (secure computer systems, encryption etc.) to protect your personal data.

Your submitted data will be held for use in the study paper, I will make sure your data is always protected and not traceable to you in any way.

The data that will be gathered through the questionnaire will not be shared with anybody outside of the UK.

What are your choices about how your information is used?
You can stop being part of the study at any time, without giving a reason, your data will be deleted if requested as well. However, once you **submit your response and data to the questionnaire, it cannot be withdrawn as it cannot be tracked back to you.**

While if any personally identifiable data is gathered, you have the right to ask us to remove, edit or delete the data gathered. However, due to the anonymity of the data collected through the questionnaire, and the lack of **personally identifiable data** within the questionnaire, once **you submit your response to the questionnaire, it cannot be withdrawn**:
https://www.port.ac.uk/about-us/structure-and-governance/legal/data-protection-and-gdpr/requesting-your-data

13. What will happen if I don't want to carry on with the study?

As a volunteer you can stop any participation at any time when filling out the questionnaire, however you **cannot withdraw your response to the questionnaire once submitted.** As there is no personally identifiable information gathered through the survey and it cannot be tracked back to you.

14. What if there is a problem?

If you have a query, concern or complaint about any aspect of this study, in the first instance you should contact the researcher(s) if appropriate. If the researcher is a student, there will also be an academic member of staff listed as the supervisor whom you can contact. If there is a complaint and there is a supervisor listed, please contact the Supervisor with details of the complaint. The contact details for both the researcher and any supervisor are detailed on page 1.

If your concern or complaint is not resolved by the researcher or their supervisor, you should contact the Head of Department:

The Head of Department	
Department / School of…..	Computing, Mathematics and Physics
University of Portsmouth	rich.boakes@port.ac.uk
Buckingham Building Portsmouth	
PO1 3HE	

If the complaint remains unresolved, please contact:

The University Complaints Officer

023 9284 3642 complaintsadvice@port.ac.uk

15. Who is funding the research?

This research isn't receiving any additional funds, none of the researchers or students managing the project will receive any financial reward by conducting this study.

16. Who has reviewed the study?

Research involving human participants is reviewed by an ethics committee to ensure that the dignity and well-being of participants is respected. This study has been reviewed by TETHIC-2026-113126 and been given favourable ethical opinion.

Thank you

Thank you for taking time to read this information sheet and for considering volunteering for this research.

Participant Information Sheet Version 1.1 - February 2026